

01_Indian_Agriculture_Crop_Production

July 9, 2025

1 Agricultural Crop Production Analysis (India): 2001- 2020

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Dataset: Government agricultural crop production data across Indian states, districts, and years.

```
[202]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[203]: agri_df = pd.read_csv('India Agriculture Crop Production.csv')
agri_df
agri_df.head()
#agri_df.tail()
```

```
[203]:
```

	State	District	Crop	Year	\
0	Andaman and Nicobar Islands	NICOBARS	Arecanut	2001-02	
1	Andaman and Nicobar Islands	NICOBARS	Arecanut	2002-03	
2	Andaman and Nicobar Islands	NICOBARS	Arecanut	2003-04	
3	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAMAN	Arecanut	2001-02	
4	Andaman and Nicobar Islands	SOUTH ANDAMANS	Arecanut	2002-03	

	Season	Area	Area Units	Production	Production Units	Yield
0	Kharif	1254.0	Hectare	2061.0	Tonnes	1.643541
1	Whole Year	1258.0	Hectare	2083.0	Tonnes	1.655803
2	Whole Year	1261.0	Hectare	1525.0	Tonnes	1.209358
3	Kharif	3100.0	Hectare	5239.0	Tonnes	1.690000
4	Whole Year	3105.0	Hectare	5267.0	Tonnes	1.696296

```
[204]: agri_df.info() # here there are null values we need to take care
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 345407 entries, 0 to 345406
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   State               345407 non-null object
1   District            345407 non-null object
```

```

2   Crop                345375 non-null object
3   Year                345407 non-null object
4   Season              345406 non-null object
5   Area                345374 non-null float64
6   Area Units          345407 non-null object
7   Production          340414 non-null float64
8   Production Units    345407 non-null object
9   Yield               345374 non-null float64
dtypes: float64(3), object(7)
memory usage: 26.4+ MB

```

```
[205]: agri_df.isna().sum() # count how many null values are there in each col.
```

```

[205]: State                0
District                  0
Crop                     32
Year                     0
Season                   1
Area                     33
Area Units               0
Production               4993
Production Units         0
Yield                   33
dtype: int64

```

```

[206]: # Since we have approx 3 lakh data and 6000 datas are null, we can remove those
agri_df = agri_df.dropna() # it will drop all the rows which have null values
      ↪by default
agri_df

```

```

[206]:
0      Andaman and Nicobar Islands  NICOBARS  Arecanut \
1      Andaman and Nicobar Islands  NICOBARS  Arecanut
2      Andaman and Nicobar Islands  NICOBARS  Arecanut
3      Andaman and Nicobar Islands  NORTH AND MIDDLE ANDAMAN  Arecanut
4      Andaman and Nicobar Islands  SOUTH ANDAMANS  Arecanut
...
345370      West Bengal  PURBA BARDHAMAN  Wheat
345371      West Bengal  PURULIA  Wheat
345372      West Bengal  PURULIA  Wheat
345373      West Bengal  PURULIA  Wheat
345374      West Bengal  PURULIA  Wheat

      Year      Season      Area Area Units  Production Production Units \
0      2001-02      Kharif  1254.0      Hectare      2061.0      Tonnes
1      2002-03  Whole Year  1258.0      Hectare      2083.0      Tonnes
2      2003-04  Whole Year  1261.0      Hectare      1525.0      Tonnes

```

3	2001-02	Kharif	3100.0	Hectare	5239.0	Tonnes
4	2002-03	Whole Year	3105.0	Hectare	5267.0	Tonnes
...	...	...	...	...	...	...
345370	2000-01	Rabi	6310.0	Hectare	15280.0	Tonnes
345371	1997-98	Rabi	1895.0	Hectare	2760.0	Tonnes
345372	1998-99	Rabi	3736.0	Hectare	5530.0	Tonnes
345373	1999-00	Rabi	2752.0	Hectare	6928.0	Tonnes
345374	2000-01	Rabi	2979.0	Hectare	7430.0	Tonnes

	Yield
0	1.643541
1	1.655803
2	1.209358
3	1.690000
4	1.696296
...	...
345370	2.421553
345371	1.456464
345372	1.480193
345373	2.517442
345374	2.494126

[340414 rows x 10 columns]

1.0.1 Unique count and values for crops, states, Districts

```
[207]: # How many number of crops we have?
agri_df['Crop'].unique()
```

```
[207]: array(['Arecanut', 'Banana', 'Black pepper', 'Cashewnut', 'Coconut',
        'Dry chillies', 'Ginger', 'Other Kharif pulses', 'other oilseeds',
        'Rice', 'Sugarcane', 'Sweet potato', 'Arhar/Tur', 'Bajra',
        'Castor seed', 'Coriander', 'Cotton(lint)', 'Gram', 'Groundnut',
        'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta',
        'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses',
        'Potato', 'Ragi', 'Rapeseed &Mustard', 'Safflower', 'Sesamum',
        'Small millets', 'Soyabean', 'Sunflower', 'Tapioca', 'Tobacco',
        'Turmeric', 'Urad', 'Wheat', 'Oilseeds total', 'Jute', 'Masoor',
        'Peas & beans (Pulses)', 'Barley', 'Garlic', 'Khesari', 'Sannhamp',
        'Guar seed', 'Moth', 'Cardamom', 'Other Cereals', 'Cowpea(Lobia)',
        'Dry Ginger', 'Other Summer Pulses'], dtype=object)
```

```
[208]: len(agri_df['Crop'].unique())
```

```
[208]: 56
```

```
[209]: print(f"There are {len(agri_df['Crop'].unique())} unique crops an they are as_
↳ follows:\n {agri_df['Crop'].unique()}")
# print(f'There are {len(agri_df[\ 'Crop\'].unique())} unique crops and they are_
↳ as follows:\n{agri_df[\ 'Crop\'].unique()}')
```

There are 56 unique crops an they are as follows:

```
['Arecanut' 'Banana' 'Black pepper' 'Cashewnut' 'Coconut' 'Dry chillies'
'Ginger' 'Other Kharif pulses' 'other oilseeds' 'Rice' 'Sugarcane'
'Sweet potato' 'Arhar/Tur' 'Bajra' 'Castor seed' 'Coriander'
'Cotton(lint)' 'Gram' 'Groundnut' 'Horse-gram' 'Jowar' 'Linseed' 'Maize'
'Mesta' 'Moong(Green Gram)' 'Niger seed' 'Onion' 'Other Rabi pulses'
'Potato' 'Ragi' 'Rapeseed &Mustard' 'Safflower' 'Sesamum' 'Small millets'
'Soyabean' 'Sunflower' 'Tapioca' 'Tobacco' 'Turmeric' 'Urad' 'Wheat'
'Oilseeds total' 'Jute' 'Masoor' 'Peas & beans (Pulses)' 'Barley'
'Garlic' 'Khesari' 'Sannhamp' 'Guar seed' 'Moth' 'Cardamom'
'Other Cereals' 'Cowpea(Lobia)' 'Dry Ginger' 'Other Summer Pulses']
```

```
[210]: agri_df['State'].unique()
```

```
[210]: array(['Andaman and Nicobar Islands', 'Andhra Pradesh',
'Arunachal Pradesh', 'Assam', 'Bihar', 'Chandigarh',
'Chhattisgarh', 'Dadra and Nagar Haveli', 'Daman and Diu', 'Delhi',
'Goa', 'Gujarat', 'Haryana', 'Himachal Pradesh',
'Jammu and Kashmir', 'Jharkhand', 'Karnataka', 'Kerala',
'Madhya Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya', 'Mizoram',
'Nagaland', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan',
'Sikkim', 'Tamil Nadu', 'Tripura', 'Uttar Pradesh', 'Uttarakhand',
'West Bengal', 'Telangana', 'Laddakh'], dtype=object)
```

```
[211]: agri_df['District'].unique()
```

```
[211]: array(['NICOBARS', 'NORTH AND MIDDLE ANDAMAN', 'SOUTH ANDAMANS',
'ANANTAPUR', 'EAST GODAVARI', 'KRISHNA', 'VIZIANAGARAM',
'WEST GODAVARI', 'ADILABAD', 'CHITTOOR', 'GUNTUR', 'KADAPA',
'KARIMNAGAR', 'KHAMMAM', 'KURNOOL', 'MAHBUBNAGAR', 'MEDAK',
'NALGONDA', 'NIZAMABAD', 'PRAKASAM', 'RANGAREDDI', 'SPSR NELLORE',
'SRIKAKULAM', 'VISAKHAPATANAM', 'WARANGAL', 'CHANGLANG',
'DIBANG VALLEY', 'EAST KAMENG', 'EAST SIANG', 'KURUNG KUMEY',
'LOHIT', 'LOWER DIBANG VALLEY', 'LOWER SUBANSIRI', 'PAPUM PARE',
'TAWANG', 'TIRAP', 'UPPER SIANG', 'UPPER SUBANSIRI', 'WEST KAMENG',
'WEST SIANG', 'BARPETA', 'BONGAIGAON', 'CACHAR', 'DARRANG',
'DHEMAJI', 'DHUBRI', 'DIBRUGARH', 'DIMA HASAO', 'GOALPARA',
'GOLAGHAT', 'HAILAKANDI', 'JORHAT', 'KAMRUP', 'KARBI ANGLONG',
'KARIMGANJ', 'KOKRAJHAR', 'LAKHIMPUR', 'MARIGAON', 'NAGAON',
'NALBARI', 'SIVASAGAR', 'SONITPUR', 'TINSUKIA', 'ARARIA', 'ARWAL',
'AURANGABAD', 'BANKA', 'BEGUSARAI', 'BHAGALPUR', 'BHOJPUR',
'BUXAR', 'DARBHANGA', 'GAYA', 'GOPALGANJ', 'JAMUI', 'JEHANABAD',
```

'KAIMUR (BHABUA)', 'KATIHAR', 'KHAGARIA', 'KISHANGANJ',
 'LAKHISARAI', 'MADHEPURA', 'MADHUBANI', 'MUNGER', 'MUZAFFARPUR',
 'NALANDA', 'NAWADA', 'PASHCHIM CHAMPARAN', 'PATNA',
 'PURBI CHAMPARAN', 'PURNIA', 'ROHTAS', 'SAHARSA', 'SAMASTIPUR',
 'SARAN', 'SHEIKHPURA', 'SHEOHAR', 'SITAMARHI', 'SIWAN', 'SUPAUL',
 'VAISHALI', 'CHANDIGARH', 'BASTAR', 'BILASPUR', 'DANTEWADA',
 'DHAMTARI', 'DURG', 'JANJGIR-CHAMPA', 'JASHPUR', 'KABIRDHAM',
 'KANKER', 'KORBA', 'KOREA', 'MAHASAMUND', 'RAIGARH', 'RAIPUR',
 'RAJNANDGAON', 'SURGUJA', 'DADRA AND NAGAR HAVELI',
 'Daman and Diu', 'Delhi', 'Goa', 'AHMADABAD', 'AMRELI', 'ANAND',
 'BANAS KANTHA', 'BHARUCH', 'BHAVNAGAR', 'DANG', 'DOHAD',
 'GANDHINAGAR', 'JAMNAGAR', 'JUNAGADH', 'KHEDA', 'MAHESANA',
 'NARMADA', 'NAVSARI', 'PANCH MAHALS', 'PATAN', 'PORBANDAR',
 'RAJKOT', 'SABAR KANTHA', 'SURAT', 'SURENDRANAGAR', 'VADODARA',
 'VALSAD', 'KACHCHH', 'AMBALA', 'BHIWANI', 'FARIDABAD', 'FATEHABAD',
 'GURGAON', 'HISAR', 'JHAJJAR', 'JIND', 'KAITHAL', 'KARNAL',
 'KURUKSHETRA', 'MAHENDRAGARH', 'PANCHKULA', 'PANIPAT', 'REWARI',
 'ROHTAK', 'SIRSA', 'SONIPAT', 'YAMUNANAGAR', 'KANGRA', 'KULLU',
 'MANDI', 'SHIMLA', 'SOLAN', 'UNA', 'CHAMBA', 'HAMIRPUR', 'SIRMAUR',
 'KINNAUR', 'LAHUL AND SPITI', 'DODA', 'JAMMU', 'KATHUA', 'RAJAURI',
 'UDHAMPUR', 'KARGIL', 'LEH LADAKH', 'SRINAGAR', 'BADGAM',
 'BARAMULLA', 'POONCH', 'PULWAMA', 'ANANTNAG', 'KUPWARA', 'CHATRA',
 'DUMKA', 'GARHWA', 'GODDA', 'GUMLA', 'HAZARIBAGH', 'KODERMA',
 'LATEHAR', 'LOHARDAGA', 'PAKUR', 'PALAMU', 'RANCHI', 'SAHEBGANJ',
 'SARAIKELA KHARSAWAN', 'SIMDEGA', 'WEST SINGHBHUM', 'BOKARO',
 'DEOGHAR', 'DHANBAD', 'EAST SINGHBUM', 'GIRIDIH', 'JAMTARA',
 'BAGALKOT', 'BANGALORE RURAL', 'BELGAUM', 'BELLARY',
 'BENGALURU URBAN', 'CHAMARAJANAGAR', 'CHIKMAGALUR', 'CHITRADURGA',
 'DAKSHIN KANNAD', 'DAVANGERE', 'DHARWAD', 'GADAG', 'HASSAN',
 'HAVERI', 'KODAGU', 'KOLAR', 'MANDYA', 'MYSORE', 'SHIMOGA',
 'TUMKUR', 'UDUPI', 'UTTAR KANNAD', 'BIDAR', 'BIJAPUR', 'GULBARGA',
 'KOPPAL', 'RAICHUR', 'ALAPPUZHA', 'ERNAKULAM', 'IDUKKI', 'KANNUR',
 'KASARAGOD', 'KOLLAM', 'KOTTAYAM', 'KOZHIKODE', 'MALAPPURAM',
 'PALAKKAD', 'PATHANAMTHITTA', 'THIRUVANANTHAPURAM', 'THRISSUR',
 'WAYANAD', 'ANUPPUR', 'ASHOKNAGAR', 'BALAGHAT', 'BARWANI', 'BETUL',
 'BHIND', 'BHOPAL', 'BURHANPUR', 'CHHATARPUR', 'CHHINDWARA',
 'DAMOH', 'DATIA', 'DEWAS', 'DHAR', 'DINDORI', 'GUNA', 'GWALIOR',
 'HARDA', 'HOSHANGABAD', 'INDORE', 'JABALPUR', 'JHABUA', 'KATNI',
 'KHANDWA', 'KHARGONE', 'MANDLA', 'MANDSAUR', 'MORENA',
 'NARSINGHPUR', 'NEEMUCH', 'PANNA', 'RAISEN', 'RAJGARH', 'RATLAM',
 'REWA', 'SAGAR', 'SATNA', 'SEHORE', 'SEONI', 'SHAHNOL', 'SHAJAPUR',
 'SHEOPUR', 'SHIVPURI', 'SIDHI', 'TIKAMGARH', 'UJJAIN', 'UMARIA',
 'VIDISHA', 'AHMEDNAGAR', 'AKOLA', 'AMRAVATI', 'BEED', 'BHANDARA',
 'BULDHANA', 'CHANDRAPUR', 'DHULE', 'GADCHIROLI', 'GONDIA',
 'HINGOLI', 'JALGAON', 'JALNA', 'KOLHAPUR', 'LATUR', 'NAGPUR',
 'NANDED', 'NANDURBAR', 'NASHIK', 'OSMANABAD', 'PARBHANI', 'PUNE',
 'RAIGAD', 'RATNAGIRI', 'SANGLI', 'SATARA', 'SOLAPUR', 'THANE',

'WARDHA', 'WASHIM', 'YAVATMAL', 'SINDHUDURG', 'SENAPATI',
 'BISHNUPUR', 'CHANDEL', 'CHURACHANDPUR', 'IMPHAL EAST',
 'IMPHAL WEST', 'TAMENGLONG', 'THOUBAL', 'UKHRUL',
 'EAST GARO HILLS', 'EAST JAINTIA HILLS', 'EAST KHASI HILLS',
 'RI BHOI', 'SOUTH GARO HILLS', 'WEST GARO HILLS',
 'WEST KHASI HILLS', 'AIZAWL', 'CHAMPHAI', 'KOLASIB', 'LUNGLEI',
 'MAMIT', 'SAIHA', 'DIMAPUR', 'KOHIMA', 'MOKOKCHUNG', 'MON', 'PHEK',
 'TUENSANG', 'WOKHA', 'ZUNHEBOTO', 'ANUGUL', 'BALANGIR',
 'BALESHWAR', 'BARGARH', 'BHADRAK', 'BOUDH', 'CUTTACK', 'DEOGARH',
 'DHENKANAL', 'GAJAPATI', 'GANJAM', 'JAGATSINGHAPUR', 'JAJAPUR',
 'JHARSUGUDA', 'KALAHANDI', 'KANDHAMAL', 'KENDRAPARA', 'KENDUJHAR',
 'KHORDHA', 'KORAPUT', 'MALKANGIRI', 'MAYURBHANJ', 'NABARANGPUR',
 'NAYAGARH', 'NUAPADA', 'RAYAGADA', 'SAMBALPUR', 'SONEPUR',
 'SUNDARGARH', 'PURI', 'MAHE', 'PONDICHERRY', 'KARAIKAL', 'YANAM',
 'AMRITSAR', 'BATHINDA', 'FARIDKOT', 'FATEHGARH SAHIB', 'FIROZEPUR',
 'HOSHIARPUR', 'JALANDHAR', 'KAPURTHALA', 'LUDHIANA', 'MOGA',
 'MUKTSAR', 'NAWANSHAHR', 'PATIALA', 'RUPNAGAR', 'SANGRUR', 'MANSA',
 'GURDASPUR', 'AJMER', 'ALWAR', 'BANSWARA', 'BARAN', 'BHARATPUR',
 'BHILWARA', 'BIKANER', 'BUNDI', 'CHITTORGARH', 'DAUSA', 'DHOLPUR',
 'DUNGARPUR', 'GANGANAGAR', 'HANUMANGARH', 'JAIPUR', 'JAISALMER',
 'JHALAWAR', 'KARALI', 'KOTA', 'NAGPUR', 'PALI', 'RAJSAMAND',
 'SAWAI MADHOPUR', 'SIKAR', 'SIROHI', 'TONK', 'UDAIPUR', 'BARMER',
 'CHURU', 'JALORE', 'JHUNJHUNU', 'JODHPUR', 'EAST DISTRICT',
 'NORTH DISTRICT', 'SOUTH DISTRICT', 'WEST DISTRICT', 'COIMBATORE',
 'CUDDALORE', 'DHARMAPURI', 'DINDIGUL', 'ERODE', 'KANCHIPURAM',
 'KARUR', 'MADURAI', 'NAMAKKAL', 'PERAMBALUR', 'PUDUKKOTTAI',
 'RAMANATHAPURAM', 'SALEM', 'SIVAGANGA', 'THANJAVUR', 'THENI',
 'THIRUVALLUR', 'THIRUVARUR', 'THOOTHUKUDI', 'TIRUCHIRAPPALLI',
 'TIRUNELVELI', 'TIRUVANNAMALAI', 'VELLORE', 'VILLUPURAM',
 'VIRUDHUNAGAR', 'NAGAPATTINAM', 'THE NILGIRIS', 'KANNIYAKUMARI',
 'KRISHNAGIRI', 'DHALAI', 'NORTH TRIPURA', 'SOUTH TRIPURA',
 'WEST TRIPURA', 'AGRA', 'ALIGARH', 'ALLAHABAD', 'AMBEDKAR NAGAR',
 'AMROHA', 'AURAIYA', 'AZAMGARH', 'BAGHPAT', 'BAHRAICH', 'BALLIA',
 'BALRAMPUR', 'BANDA', 'BARABANKI', 'BAREILLY', 'BASTI', 'BIJNOR',
 'BUDAUN', 'BULANDSHAHR', 'CHANDAULI', 'CHITRAKOOT', 'DEORIA',
 'ETAH', 'ETAWAH', 'FAIZABAD', 'FARRUKHABAD', 'FATEHPUR',
 'FIROZABAD', 'GAUTAM BUDDHA NAGAR', 'GHAZIABAD', 'GHAZIPUR',
 'GONDA', 'GORAKHPUR', 'HARDOI', 'HATHRAS', 'JALAUN', 'JAUNPUR',
 'JHANSI', 'KANNAUJ', 'KANPUR DEHAT', 'KANPUR NAGAR', 'KAUSHAMBI',
 'KHERI', 'KUSHI NAGAR', 'LALITPUR', 'LUCKNOW', 'MAHARAJGANJ',
 'MAHOBA', 'MAINPURI', 'MATHURA', 'MAU', 'MEERUT', 'MIRZAPUR',
 'MORADABAD', 'MUZAFFARNAGAR', 'PILIBHIT', 'PRATAPGARH',
 'RAE BARELI', 'RAMPUR', 'SAHARANPUR', 'SANT KABEER NAGAR',
 'SANT RAVIDAS NAGAR', 'SHAHJAHANPUR', 'SHRAVASTI',
 'SIDDHARTH NAGAR', 'SITAPUR', 'SONBHADRA', 'SULTANPUR', 'UNNAO',
 'VARANASI', 'CHAMOLI', 'CHAMPAWAT', 'DEHRADUN', 'HARIDWAR',
 'NAINITAL', 'PAURI GARHWAL', 'PITHORAGARH', 'RUDRA PRAYAG',

```

'TEHRI GARHWAL', 'UDAM SINGH NAGAR', 'UTTAR KASHI', 'ALMORA',
'BAGESHWAR', '24 PARAGANAS NORTH', '24 PARAGANAS SOUTH', 'BANKURA',
'BIRBHUM', 'COOCHBEHAR', 'DARJEELING', 'DINAJPUR DAKSHIN',
'DINAJPUR UTTAR', 'HOOGHLY', 'HOWRAH', 'JALPAIGURI', 'MALDAH',
'MEDINIPUR EAST', 'MEDINIPUR WEST', 'MURSHIDABAD', 'NADIA',
'PURBA BARDHAMAN', 'PURULIA', 'ANJAW', 'BAKSA', 'CHIRANG',
'KAMRUP METRO', 'UDALGURI', 'NARAYANPUR', 'NORTH GOA', 'SOUTH GOA',
'MEWAT', 'PALGHAR', 'LAWNGTLAI', 'SERCHHIP', 'S', 'TARN TARAN',
'Andaman and Nicobar Islands', 'DELHI_TOTAL', 'TAPI', 'PALWAL',
'REASI', 'SAMBA', 'KISHTWAR', 'RAMBAN', 'KULGAM', 'BANDIPORA',
'GANDERBAL', 'SHOPIAN', 'KHUNTI', 'RAMGARH', 'RAMANAGARA',
'CHIKBALLAPUR', 'ALIRAJPUR', 'SINGRAULI', 'KIPHIRE', 'LONGLENG',
'PEREN', 'BARNALA', 'TIRUPPUR', 'ARIYALUR', 'KASGANJ', 'BALOD',
'BALODA BAZAR', 'BEMETARA', 'GARIYABAND', 'KONDAGAON', 'MUNGELI',
'SUKMA', 'SURAJPUR', 'YADGIR', 'FAZILKA', 'PATHANKOT', 'GOMATI',
'KHOWAI', 'SEPAHIJALA', 'UNAKOTI', 'AMETHI', 'HAPUR', 'SAMBHAL',
'SHAMLI', 'KRA DAADI', 'LONGDING', 'NAMSAI', 'SIANG', 'DIU',
'DAMAN', 'ARAVALLI', 'BOTAD', 'CHHOTAUDEPUR', 'GIR SOMNATH',
'MAHISAGAR', 'DEVBHUMI DWARKA', 'MORBI', 'CHARKI DADRI',
'AGAR MALWA', 'NORTH GARO HILLS', 'SOUTH WEST GARO HILLS',
'SOUTH WEST KHASI HILLS', 'WEST JAINTIA HILLS', 'ALIPURDUAR',
'KAMLE', 'LOWER SIANG', 'MUMBAI SUBURBAN', 'BHADRADRI', 'JAGITIAL',
'JANGOAN', 'JAYASHANKAR', 'JOGULAMBA', 'KAMAREDDY',
'KOMARAM BHEEM ASIFABAD', 'MAHABUBABAD', 'MANCHERIAL', 'MEDCHAL',
'MULUGU', 'NAGARKURNOOL', 'NARAYANAPET', 'NIRMAL', 'PEDDAPALLI',
'RAJANNA', 'SANGAREDDY', 'SIDDIPET', 'SURYAPET', 'VIKARABAD',
'WANAPARTHY', 'WARANGAL URBAN', 'YADADRI', 'JHARGRAM',
'PASCHIM BARDHAMAN', 'KALIMPONG', 'LEPARADA', 'PAKKE KESSANG',
'SHI YOMI', 'BISWANATH', 'CHARAIDEO', 'HOJAI', 'MAJULI',
'SOUTH SALMARA MANCACHAR', 'WEST KARBI ANGLONG',
'GAURELLA-PENDRA-MARWAHI', 'BAGALKOTE', 'BALLARI', 'BELAGAVI',
'CHAMARAJANAGARA', 'CHIKKABALLAPURA', 'CHIKKAMAGALURU',
'DAKSHINA KANNADA', 'MYSURU', 'SHIVAMOGGA', 'TUMAKURU',
'UTTARA KANNADA', 'VIJAYAPURA', 'KALABURAGI', 'YADAGIRI', 'NIWARI',
'SHAHID BHAGAT SINGH NAGAR', 'TUTICORIN', 'CHENNAI',
'KALLAKURICHI', 'TENKASI', 'TIRUPATHUR', 'CHENGALPATTU', 'RANIPET',
'HANUMAKONDA', 'MEDCHAL MALKAJGIRI', 'HYDERABAD', 'MUMBAI'],
dtype=object)

```

```

[212]: print(f"Records are available for {len(agri_df['District'].unique())} districts_
        ↪in {len(agri_df['State'].unique())} states")

```

Records are available for 728 districts in 36 states

1.0.2 Dealing with different production Units:

```
[213]: agri_df['Production Units'].value_counts()
```

```
[213]: Tonnes      326999
      Bales       10524
      Nuts         2891
      Name: Production Units, dtype: int64
```

```
[214]: # Assume one cocunut is 2kg
      agri_df['Production Units'] == 'Nuts'
```

```
[214]: 0      False
      1      False
      2      False
      3      False
      4      False
      ...
      345370  False
      345371  False
      345372  False
      345373  False
      345374  False
      Name: Production Units, Length: 340414, dtype: bool
```

```
[ ]:
```

```
[215]: agri_df[agri_df['Production Units'] == 'Nuts']
```

```
[215]:
```

	State	District	Crop \
20	Andaman and Nicobar Islands	NICOBARS	Coconut
21	Andaman and Nicobar Islands	NICOBARS	Coconut
22	Andaman and Nicobar Islands	NICOBARS	Coconut
23	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAMAN	Coconut
24	Andaman and Nicobar Islands	SOUTH ANDAMANS	Coconut
...	...	...	...
343479	West Bengal	PURBA BARDHAMAN	Coconut
343480	West Bengal	PURULIA	Coconut
343481	West Bengal	PURULIA	Coconut
343482	West Bengal	PURULIA	Coconut
343483	West Bengal	PURULIA	Coconut

	Year	Season	Area	Area Units	Production	Production Units \
20	2001-02	Whole Year	18190.00	Hectare	64430000.0	Nuts
21	2002-03	Whole Year	18240.00	Hectare	67490000.0	Nuts
22	2003-04	Whole Year	18284.74	Hectare	68580000.0	Nuts
23	2001-02	Whole Year	7015.00	Hectare	25250000.0	Nuts

24	2002-03	Whole Year	7060.00	Hectare	26830000.0	Nuts
...	...	...	...	...	...	...
343479	2000-01	Whole Year	1952.00	Hectare	28460200.0	Nuts
343480	1997-98	Whole Year	17.00	Hectare	340000.0	Nuts
343481	1998-99	Whole Year	20.00	Hectare	310000.0	Nuts
343482	1999-00	Whole Year	20.00	Hectare	290000.0	Nuts
343483	2000-01	Whole Year	27.00	Hectare	364500.0	Nuts

	Yield
20	3542.056075
21	3700.109649
22	3750.668590
23	3599.429793
24	3800.283286
...	...
343479	14580.020492
343480	20000.000000
343481	15500.000000
343482	14500.000000
343483	13500.000000

[2891 rows x 10 columns]

```
[216]: agri_df[agri_df['Production Units'] == 'Nuts']['Crop']
```

```
[216]: 20      Coconut
      21      Coconut
      22      Coconut
      23      Coconut
      24      Coconut
      ...
      343479  Coconut
      343480  Coconut
      343481  Coconut
      343482  Coconut
      343483  Coconut
      Name: Crop, Length: 2891, dtype: object
```

```
[217]: agri_df[agri_df['Production Units'] == 'Nuts']['Crop'].value_counts()
```

```
[217]: Coconut      2891
      Name: Crop, dtype: int64
```

So nuts units is used only for cocunut here. (from above)

```
[218]: # Convert the nuts to kgs
      def unit_conversion(df):
```

```

if df['Production Units'] == 'Nuts':
    return df['Production']/50
elif df['Production Units'] == 'Tonnes':
    return df['Production']
else:
    return df['Production']/4.166

```

```
[219]: agri_df['new production'] = np.round(agri_df.apply(unit_conversion,axis = 1),2)
```

```
[220]: #agri_df = agri_df.drop('ne production', axis=1)
```

```
[221]: agri_df.sample(15)
```

```
[221]:
```

	State	District	Crop	Year \
164828	Rajasthan	BHILWARA	Peas & beans (Pulses)	2012-13
90261	Bihar	ARWAL	Arhar/Tur	2009-10
204851	Nagaland	MON	Gram	2015-16
257947	Odisha	SAMBALPUR	Rice	2016-17
245730	Karnataka	MYSORE	Rice	2017-18
135293	Assam	GOALPARA	Tapioca	2012-13
269035	Uttar Pradesh	AMETHI	Arhar/Tur	2017-18
2764	Arunachal Pradesh	LOWER SUBANSIRI	Ginger	2001-02
103536	Karnataka	UDUPI	Rice	2009-10
145278	Jammu and Kashmir	KATHUA	Other Rabi pulses	2011-12
306995	Bihar	KISHANGANJ	Ragi	2000-01
2291	Andhra Pradesh	ANANTAPUR	Sunflower	2002-03
145870	Jharkhand	CHATRA	Gram	2011-12
111517	Maharashtra	BULDHANA	Wheat	2009-10
146035	Jharkhand	GARHWA	Onion	2010-11

	Season	Area	Area Units	Production	Production Units \
164828	Rabi	41.0	Hectare	4.0	Tonnes
90261	Kharif	453.0	Hectare	647.0	Tonnes
204851	Rabi	70.0	Hectare	60.0	Tonnes
257947	Winter	111940.0	Hectare	224040.0	Tonnes
245730	Summer	7951.0	Hectare	24488.0	Tonnes
135293	Whole Year	220.0	Hectare	1595.0	Tonnes
269035	Kharif	3982.0	Hectare	3021.0	Tonnes
2764	Whole Year	75.0	Hectare	530.0	Tonnes
103536	Summer	143.0	Hectare	316.0	Tonnes
145278	Rabi	2.0	Hectare	0.9	Tonnes
306995	Kharif	902.0	Hectare	2342.0	Tonnes
2291	Kharif	60319.0	Hectare	22149.0	Tonnes
145870	Rabi	1335.0	Hectare	1366.0	Tonnes
111517	Rabi	46000.0	Hectare	93900.0	Tonnes
146035	Rabi	245.0	Hectare	1365.0	Tonnes

	Yield	new production
164828	0.097561	4.0
90261	1.428256	647.0
204851	0.857143	60.0
257947	2.001429	224040.0
245730	3.079864	24488.0
135293	7.250000	1595.0
269035	0.758664	3021.0
2764	7.066667	530.0
103536	2.209790	316.0
145278	0.450000	0.9
306995	2.596452	2342.0
2291	0.367198	22149.0
145870	1.023221	1366.0
111517	2.041304	93900.0
146035	5.571429	1365.0

1.0.3 Removing the repeated info columns

```
[222]: agri_df.drop(columns = ['Production', 'Production Units'])
```

```
[222]:
```

	State	District	Crop \
0	Andaman and Nicobar Islands	NICOBARS	Arecanut
1	Andaman and Nicobar Islands	NICOBARS	Arecanut
2	Andaman and Nicobar Islands	NICOBARS	Arecanut
3	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAMAN	Arecanut
4	Andaman and Nicobar Islands	SOUTH ANDAMANS	Arecanut
...	...	...	...
345370	West Bengal	PURBA BARDHAMAN	Wheat
345371	West Bengal	PURULIA	Wheat
345372	West Bengal	PURULIA	Wheat
345373	West Bengal	PURULIA	Wheat
345374	West Bengal	PURULIA	Wheat

	Year	Season	Area	Area Units	Yield	new production
0	2001-02	Kharif	1254.0	Hectare	1.643541	2061.0
1	2002-03	Whole Year	1258.0	Hectare	1.655803	2083.0
2	2003-04	Whole Year	1261.0	Hectare	1.209358	1525.0
3	2001-02	Kharif	3100.0	Hectare	1.690000	5239.0
4	2002-03	Whole Year	3105.0	Hectare	1.696296	5267.0
...	...	...	...	...	...	...
345370	2000-01	Rabi	6310.0	Hectare	2.421553	15280.0
345371	1997-98	Rabi	1895.0	Hectare	1.456464	2760.0
345372	1998-99	Rabi	3736.0	Hectare	1.480193	5530.0
345373	1999-00	Rabi	2752.0	Hectare	2.517442	6928.0
345374	2000-01	Rabi	2979.0	Hectare	2.494126	7430.0

[340414 rows x 9 columns]

```
[223]: agri_df = agri_df.drop(columns = ['Production', 'Production Units'])
```

```
[224]: agri_df
```

```
[224]:
```

	State	District	Crop \
0	Andaman and Nicobar Islands	NICOBARS	Arecanut
1	Andaman and Nicobar Islands	NICOBARS	Arecanut
2	Andaman and Nicobar Islands	NICOBARS	Arecanut
3	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAMAN	Arecanut
4	Andaman and Nicobar Islands	SOUTH ANDAMANS	Arecanut
...	...	...	...
345370	West Bengal	PURBA BARDHAMAN	Wheat
345371	West Bengal	PURULIA	Wheat
345372	West Bengal	PURULIA	Wheat
345373	West Bengal	PURULIA	Wheat
345374	West Bengal	PURULIA	Wheat

	Year	Season	Area	Area Units	Yield	new production
0	2001-02	Kharif	1254.0	Hectare	1.643541	2061.0
1	2002-03	Whole Year	1258.0	Hectare	1.655803	2083.0
2	2003-04	Whole Year	1261.0	Hectare	1.209358	1525.0
3	2001-02	Kharif	3100.0	Hectare	1.690000	5239.0
4	2002-03	Whole Year	3105.0	Hectare	1.696296	5267.0
...	...	...	...	...	...	...
345370	2000-01	Rabi	6310.0	Hectare	2.421553	15280.0
345371	1997-98	Rabi	1895.0	Hectare	1.456464	2760.0
345372	1998-99	Rabi	3736.0	Hectare	1.480193	5530.0
345373	1999-00	Rabi	2752.0	Hectare	2.517442	6928.0
345374	2000-01	Rabi	2979.0	Hectare	2.494126	7430.0

[340414 rows x 9 columns]

```
[225]: agri_df.head()
```

```
[225]:
```

	State	District	Crop	Year \
0	Andaman and Nicobar Islands	NICOBARS	Arecanut	2001-02
1	Andaman and Nicobar Islands	NICOBARS	Arecanut	2002-03
2	Andaman and Nicobar Islands	NICOBARS	Arecanut	2003-04
3	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAMAN	Arecanut	2001-02
4	Andaman and Nicobar Islands	SOUTH ANDAMANS	Arecanut	2002-03

	Season	Area	Area Units	Yield	new production
0	Kharif	1254.0	Hectare	1.643541	2061.0
1	Whole Year	1258.0	Hectare	1.655803	2083.0

2	Whole Year	1261.0	Hectare	1.209358	1525.0
3	Kharif	3100.0	Hectare	1.690000	5239.0
4	Whole Year	3105.0	Hectare	1.696296	5267.0

```
[226]: # Common crop choice for agriculture in India
agri_df['Crop'].value_counts()
```

```
[226]: Rice                21573
       Maize                20329
       Moong(Green Gram)   14813
       Urad                 14389
       Sesamum             12744
       Groundnut           12515
       Wheat               11219
       Rapeseed &Mustard   10919
       Sugarcane           10826
       Arhar/Tur           10774
       Potato              10729
       Onion               10621
       Gram                10330
       Jowar               9682
       Dry chillies        8890
       Bajra               7740
       Peas & beans (Pulses) 7179
       Sunflower           7104
       Small millets       6859
       Cotton(lint)        6318
       Masoor              6296
       Turmeric            5916
       Barley              5808
       Linseed             5793
       Ragi                5725
       Sweet potato        5656
       Other Kharif pulses  5590
       Horse-gram          5294
       Garlic              4958
       Coriander           4954
       Soyabean            4953
       Other Rabi pulses   4728
       Ginger              4652
       Castor seed         4496
       Banana              4393
       Tobacco             3513
       Coconut             2891
       Sannhamp            2738
       Niger seed          2707
       Mesta               2369
```

```

Tapioca                2264
Arecanut               2150
Guar seed              2045
Jute                   1837
Khesari               1753
Safflower             1727
Cowpea(Lobia)         1697
Cashewnut             1519
Black pepper          1370
Other Cereals          1338
Moth                  1287
other oilseeds         1190
Oilseeds total         702
Cardamom              483
Other Summer Pulses    66
Dry Ginger             3
Name: Crop, dtype: int64

```

```
[227]: agri_df['Crop'].value_counts().head(10) # Top 10.
```

```

[227]: Rice                21573
Maize                 20329
Moong(Green Gram)    14813
Urad                 14389
Sesamum              12744
Groundnut            12515
Wheat               11219
Rapeseed &Mustard    10919
Sugarcane            10826
Arhar/Tur           10774
Name: Crop, dtype: int64

```

1.0.4 Highest producer of rice since 1997

```
[228]: agri_df[agri_df['Crop'] == 'Rice']
```

```

[228]:
   State      District  Crop  Year \
41  Andaman and Nicobar Islands  NICOBARS  Rice  2001-02
42  Andaman and Nicobar Islands  NICOBARS  Rice  2002-03
43  Andaman and Nicobar Islands  NICOBARS  Rice  2003-04
44  Andaman and Nicobar Islands  NORTH AND MIDDLE ANDAMAN  Rice  2001-02
45  Andaman and Nicobar Islands  SOUTH ANDAMANS  Rice  2002-03
...
344825  West Bengal  PURULIA  Rice  1999-00
344826  West Bengal  PURULIA  Rice  1999-00
344827  West Bengal  PURULIA  Rice  2000-01
344828  West Bengal  PURULIA  Rice  2000-01

```

344829 West Bengal PURULIA Rice 2000-01

	Season	Area	Area Units	Yield	new production
41	Kharif	83.0	Hectare	3.614458	300.00
42	Kharif	189.2	Hectare	2.700000	510.84
43	Kharif	52.0	Hectare	1.734038	90.17
44	Kharif	9718.0	Hectare	2.781745	27033.00
45	Kharif	10695.8	Hectare	2.954507	31600.82
...	...	...	...	...	...
344825	Summer	3576.0	Hectare	2.418904	8650.00
344826	Winter	294107.0	Hectare	1.821038	535580.00
344827	Autumn	2632.0	Hectare	0.995441	2620.00
344828	Summer	1023.0	Hectare	2.398827	2454.00
344829	Winter	258531.0	Hectare	1.817616	469910.00

[21573 rows x 9 columns]

```
[229]: agri_rice = agri_df[agri_df['Crop'] == 'Rice']
agri_rice['State'].unique()
```

```
[229]: array(['Andaman and Nicobar Islands', 'Andhra Pradesh',
        'Arunachal Pradesh', 'Assam', 'Bihar', 'Chandigarh',
        'Chhattisgarh', 'Dadra and Nagar Haveli', 'Daman and Diu', 'Delhi',
        'Goa', 'Gujarat', 'Haryana', 'Himachal Pradesh',
        'Jammu and Kashmir', 'Jharkhand', 'Karnataka', 'Kerala',
        'Madhya Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya', 'Mizoram',
        'Nagaland', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan',
        'Sikkim', 'Tamil Nadu', 'Tripura', 'Uttar Pradesh', 'Uttarakhand',
        'West Bengal', 'Telangana'], dtype=object)
```

```
[230]: unique_rice_growing_states = agri_rice['State'].unique()
```

```
[231]: total_rice_production = []
for state in unique_rice_growing_states:
    #print(agri_rice.loc[agri_rice['State'] == state, 'new production'].sum())
    total_rice_production.append(agri_rice.loc[agri_rice['State'] == state,
↪ 'new production'].sum())
```

```
[232]: total_rice_production
```

```
[232]: [402806.13,
        239361201.0,
        3851550.4,
        97989518.0,
        131489958.0,
        4960.0,
        116274060.0,
```

```

585118.6,
49050.380000000005,
391753.0,
2817936.0,
32098665.0,
80531900.0,
2514409.0,
9853942.29,
28771595.53,
82049317.78,
13284212.61,
64509646.97,
61238128.0,
9426598.0,
5169434.0,
1719533.7000000002,
7818760.0,
153257182.0,
1260683.0,
243042000.0,
6046961.0,
474991.0,
132142328.0,
14710039.0,
296108280.0,
12568071.0,
338984869.0,
45628714.0]

```

```

[233]: rice_crop_production_statewise = pd.DataFrame({"State":
↳unique_rice_growing_states,
"Total Rice Production" : total_rice_production})
rice_crop_production_statewise

```

```

[233]:

```

	State	Total Rice Production
0	Andaman and Nicobar Islands	4.028061e+05
1	Andhra Pradesh	2.393612e+08
2	Arunachal Pradesh	3.851550e+06
3	Assam	9.798952e+07
4	Bihar	1.314900e+08
5	Chandigarh	4.960000e+03
6	Chhattisgarh	1.162741e+08
7	Dadra and Nagar Haveli	5.851186e+05
8	Daman and Diu	4.905038e+04
9	Delhi	3.917530e+05
10	Goa	2.817936e+06
11	Gujarat	3.209866e+07

12	Haryana	8.053190e+07
13	Himachal Pradesh	2.514409e+06
14	Jammu and Kashmir	9.853942e+06
15	Jharkhand	2.877160e+07
16	Karnataka	8.204932e+07
17	Kerala	1.328421e+07
18	Madhya Pradesh	6.450965e+07
19	Maharashtra	6.123813e+07
20	Manipur	9.426598e+06
21	Meghalaya	5.169434e+06
22	Mizoram	1.719534e+06
23	Nagaland	7.818760e+06
24	Odisha	1.532572e+08
25	Puducherry	1.260683e+06
26	Punjab	2.430420e+08
27	Rajasthan	6.046961e+06
28	Sikkim	4.749910e+05
29	Tamil Nadu	1.321423e+08
30	Tripura	1.471004e+07
31	Uttar Pradesh	2.961083e+08
32	Uttarakhand	1.256807e+07
33	West Bengal	3.389849e+08
34	Telangana	4.562871e+07

```
[234]: # I want highest one
rice_crop_production_statewise.sort_values('Total Rice Production', ascending =
↪False)
```

```
[234]:
```

	State	Total Rice Production
33	West Bengal	3.389849e+08
31	Uttar Pradesh	2.961083e+08
26	Punjab	2.430420e+08
1	Andhra Pradesh	2.393612e+08
24	Odisha	1.532572e+08
29	Tamil Nadu	1.321423e+08
4	Bihar	1.314900e+08
6	Chhattisgarh	1.162741e+08
3	Assam	9.798952e+07
16	Karnataka	8.204932e+07
12	Haryana	8.053190e+07
18	Madhya Pradesh	6.450965e+07
19	Maharashtra	6.123813e+07
34	Telangana	4.562871e+07
11	Gujarat	3.209866e+07
15	Jharkhand	2.877160e+07
30	Tripura	1.471004e+07
17	Kerala	1.328421e+07

32	Uttarakhand	1.256807e+07
14	Jammu and Kashmir	9.853942e+06
20	Manipur	9.426598e+06
23	Nagaland	7.818760e+06
27	Rajasthan	6.046961e+06
21	Meghalaya	5.169434e+06
2	Arunachal Pradesh	3.851550e+06
10	Goa	2.817936e+06
13	Himachal Pradesh	2.514409e+06
22	Mizoram	1.719534e+06
25	Puducherry	1.260683e+06
7	Dadra and Nagar Haveli	5.851186e+05
28	Sikkim	4.749910e+05
0	Andaman and Nicobar Islands	4.028061e+05
9	Delhi	3.917530e+05
8	Daman and Diu	4.905038e+04
5	Chandigarh	4.960000e+03

```
[235]: print(f"Top 10 rice producing states in (Tonnes) are: \n")
        ↪{(rice_crop_production_statewise.sort_values('Total Rice Production',
        ↪ascending = False)).head(10)}")
```

Top 10 rice producing states in (Tonnes) are:

	State	Total Rice Production
33	West Bengal	3.389849e+08
31	Uttar Pradesh	2.961083e+08
26	Punjab	2.430420e+08
1	Andhra Pradesh	2.393612e+08
24	Odisha	1.532572e+08
29	Tamil Nadu	1.321423e+08
4	Bihar	1.314900e+08
6	Chhattisgarh	1.162741e+08
3	Assam	9.798952e+07
16	Karnataka	8.204932e+07

1.0.5 Statewise total production of crops:

```
[236]: unique_states = agri_df['State'].unique()
total_production_list = []
for state in unique_states:
    total_production_list.append(agri_df.loc[agri_df['State']==state, 'new
    ↪production'].sum())

crop_production_statewise = pd.DataFrame({"State":unique_states,
                                           ↪"Total crop production":
                                           ↪total_production_list})
crop_production_statewise
```

```
[236]:
```

	State	Total crop production
0	Andaman and Nicobar Islands	4.209101e+07
1	Andhra Pradesh	1.322852e+09
2	Arunachal Pradesh	9.522010e+06
3	Assam	2.388954e+08
4	Bihar	5.221173e+08
5	Chandigarh	8.978150e+04
6	Chhattisgarh	1.430539e+08
7	Dadra and Nagar Haveli	2.022090e+06
8	Daman and Diu	5.926708e+04
9	Delhi	2.666018e+06
10	Goa	4.880315e+07
11	Gujarat	6.987092e+08
12	Haryana	5.605814e+08
13	Himachal Pradesh	3.229696e+07
14	Jammu and Kashmir	3.041253e+07
15	Jharkhand	4.379384e+07
16	Karnataka	2.316906e+09
17	Kerala	2.685621e+09
18	Madhya Pradesh	8.132793e+08
19	Maharashtra	1.792898e+09
20	Manipur	1.876435e+07
21	Meghalaya	1.521530e+07
22	Mizoram	2.762566e+06
23	Nagaland	1.859859e+07
24	Odisha	1.927792e+08
25	Puducherry	1.655736e+07
26	Punjab	7.552307e+08
27	Rajasthan	5.693465e+08
28	Sikkim	2.744927e+06
29	Tamil Nadu	2.563449e+09
30	Tripura	1.677586e+07
31	Uttar Pradesh	4.442550e+09
32	Uttarakhand	1.796973e+08
33	West Bengal	8.912222e+08
34	Telangana	1.083711e+08
35	Laddakh	1.092000e+02

1.0.6 Best year for Agriculture

```
[237]: agri_df['Year'].unique()
```

```
[237]: array(['2001-02', '2002-03', '2003-04', '2004-05', '2006-07', '2005-06',
        '2007-08', '2008-09', '2009-10', '2010-11', '2011-12', '2012-13',
        '2013-14', '2014-15', '2015-16', '2016-17', '2017-18', '2018-19',
        '2019-20', '2020-21', '2000-01', '1997-98', '1998-99', '1999-00'],
        dtype=object)
```

```
[238]: # Convert this array to list
list(agri_df['Year'].unique())
```

```
[238]: ['2001-02',
        '2002-03',
        '2003-04',
        '2004-05',
        '2006-07',
        '2005-06',
        '2007-08',
        '2008-09',
        '2009-10',
        '2010-11',
        '2011-12',
        '2012-13',
        '2013-14',
        '2014-15',
        '2015-16',
        '2016-17',
        '2017-18',
        '2018-19',
        '2019-20',
        '2020-21',
        '2000-01',
        '1997-98',
        '1998-99',
        '1999-00']
```

```
[239]: unique_year_list = list(agri_df['Year'].unique())
```

```
[240]: year_wise_production = []
for year in unique_year_list:
    year_wise_production.append(agri_df.loc[agri_df['Year'] == year, 'new_
    ↪production'].sum())

crop_production_yearwise = crop_production_yearwise = pd.DataFrame({"year":
    ↪unique_year_list,
    "Total Crop Production":year_wise_production})

# Sort the values in decending order
crop_production_yearwise.sort_values('Total Crop Production', ascending =
    ↪False).head(5)
```

```
[240]:
```

	year	Total Crop Production
18	2019-20	1.202953e+09
17	2018-19	1.180607e+09
16	2017-18	1.130183e+09

```
10 2011-12          1.091307e+09
13 2014-15          1.086199e+09
```

```
[241]: # Worse
crop_production_yearwise.sort_values('Total Crop Production', ascending = True).
      head(5)
```

```
[241]:      year  Total Crop Production
19 2020-21          1.017723e+07
21 1997-98          5.595735e+08
22 1998-99          6.848303e+08
23 1999-00          7.344804e+08
20 2000-01          7.366483e+08
```

```
[242]: agri_df['Area Units'].value_counts()
```

```
[242]: Hectare      340414
Name: Area Units, dtype: int64
```

```
[243]: agri_df
```

```
[243]:
```

	State	District	Crop \
0	Andaman and Nicobar Islands	NICOBARS	Arecanut
1	Andaman and Nicobar Islands	NICOBARS	Arecanut
2	Andaman and Nicobar Islands	NICOBARS	Arecanut
3	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAMAN	Arecanut
4	Andaman and Nicobar Islands	SOUTH ANDAMANS	Arecanut
...	...	...	...
345370	West Bengal	PURBA BARDHAMAN	Wheat
345371	West Bengal	PURULIA	Wheat
345372	West Bengal	PURULIA	Wheat
345373	West Bengal	PURULIA	Wheat
345374	West Bengal	PURULIA	Wheat

	Year	Season	Area	Area Units	Yield	new production
0	2001-02	Kharif	1254.0	Hectare	1.643541	2061.0
1	2002-03	Whole Year	1258.0	Hectare	1.655803	2083.0
2	2003-04	Whole Year	1261.0	Hectare	1.209358	1525.0
3	2001-02	Kharif	3100.0	Hectare	1.690000	5239.0
4	2002-03	Whole Year	3105.0	Hectare	1.696296	5267.0
...	...	...	...	...	...	...
345370	2000-01	Rabi	6310.0	Hectare	2.421553	15280.0
345371	1997-98	Rabi	1895.0	Hectare	1.456464	2760.0
345372	1998-99	Rabi	3736.0	Hectare	1.480193	5530.0
345373	1999-00	Rabi	2752.0	Hectare	2.517442	6928.0
345374	2000-01	Rabi	2979.0	Hectare	2.494126	7430.0

[340414 rows x 9 columns]

1.0.7 Visualization:

```
[244]: agri_df.head(5)
```

```
[244]:
```

	State	District	Crop	Year	\
0	Andaman and Nicobar Islands	NICOBARS	Arecanut	2001-02	
1	Andaman and Nicobar Islands	NICOBARS	Arecanut	2002-03	
2	Andaman and Nicobar Islands	NICOBARS	Arecanut	2003-04	
3	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAMAN	Arecanut	2001-02	
4	Andaman and Nicobar Islands	SOUTH ANDAMANS	Arecanut	2002-03	

	Season	Area	Area Units	Yield	new production
0	Kharif	1254.0	Hectare	1.643541	2061.0
1	Whole Year	1258.0	Hectare	1.655803	2083.0
2	Whole Year	1261.0	Hectare	1.209358	1525.0
3	Kharif	3100.0	Hectare	1.690000	5239.0
4	Whole Year	3105.0	Hectare	1.696296	5267.0

```
[245]: top_crops = agri_df['Crop'].value_counts().head(10)
top_crops
```

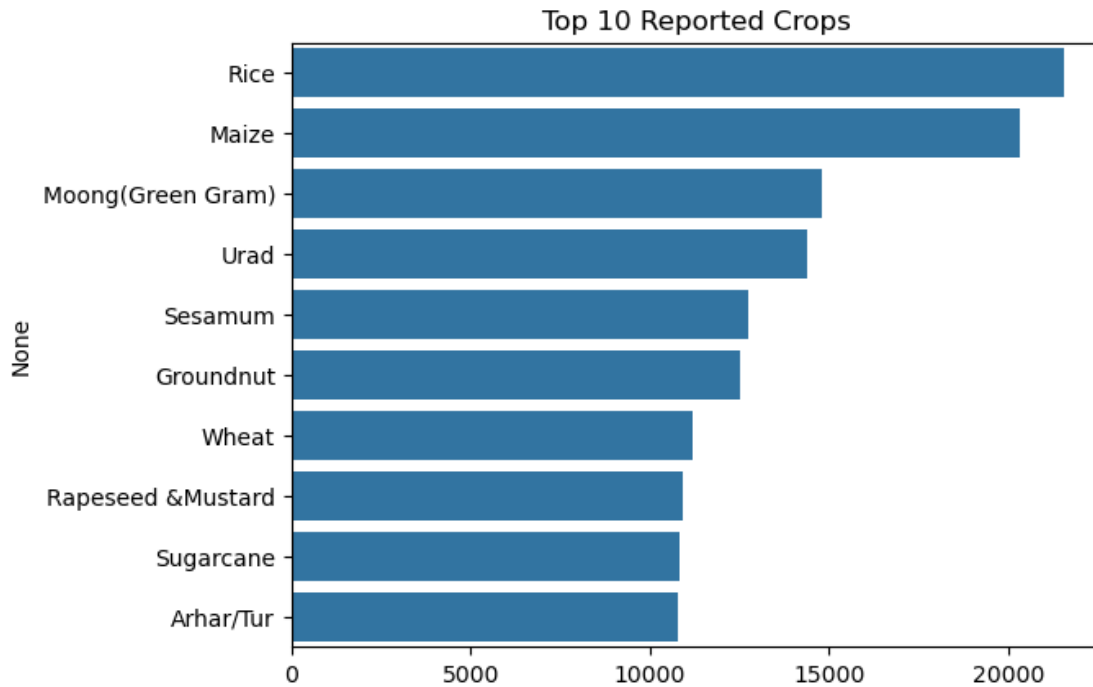
```
[245]:
```

Rice	21573
Maize	20329
Moong(Green Gram)	14813
Urad	14389
Sesamum	12744
Groundnut	12515
Wheat	11219
Rapeseed &Mustard	10919
Sugarcane	10826
Arhar/Tur	10774

Name: Crop, dtype: int64

```
[246]: # Top 10 crops
top_crops = agri_df['Crop'].value_counts().head(10)
sns.barplot(x=top_crops.values, y=top_crops.index)
plt.title("Top 10 Reported Crops")
```

```
[246]: Text(0.5, 1.0, 'Top 10 Reported Crops')
```



1.0.8 Time Series Analysis

```
[247]: agri_df[agri_df['Crop'] == 'Rice']
```

```
[247]:
```

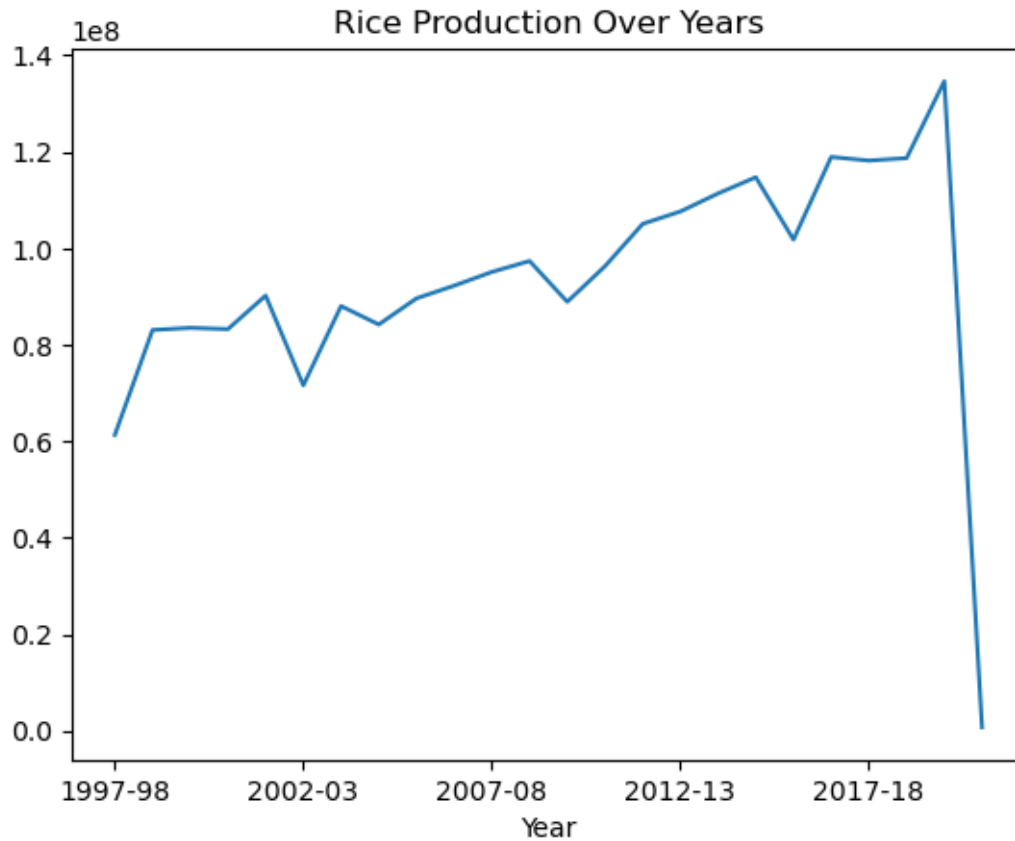
	State	District	Crop	Year	\
41	Andaman and Nicobar Islands	NICOBARS	Rice	2001-02	
42	Andaman and Nicobar Islands	NICOBARS	Rice	2002-03	
43	Andaman and Nicobar Islands	NICOBARS	Rice	2003-04	
44	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAMAN	Rice	2001-02	
45	Andaman and Nicobar Islands	SOUTH ANDAMANS	Rice	2002-03	
...	...	...	...	...	
344825	West Bengal	PURULIA	Rice	1999-00	
344826	West Bengal	PURULIA	Rice	1999-00	
344827	West Bengal	PURULIA	Rice	2000-01	
344828	West Bengal	PURULIA	Rice	2000-01	
344829	West Bengal	PURULIA	Rice	2000-01	
...	...	...	...	...	
	Season	Area	Area Units	Yield	new production
41	Kharif	83.0	Hectare	3.614458	300.00
42	Kharif	189.2	Hectare	2.700000	510.84
43	Kharif	52.0	Hectare	1.734038	90.17
44	Kharif	9718.0	Hectare	2.781745	27033.00
45	Kharif	10695.8	Hectare	2.954507	31600.82
...	...	...	...	...	...

344825	Summer	3576.0	Hectare	2.418904	8650.00
344826	Winter	294107.0	Hectare	1.821038	535580.00
344827	Autumn	2632.0	Hectare	0.995441	2620.00
344828	Summer	1023.0	Hectare	2.398827	2454.00
344829	Winter	258531.0	Hectare	1.817616	469910.00

[21573 rows x 9 columns]

```
[248]: # Total production of Rice over years
rice_data = agri_df[agri_df['Crop'] == 'Rice']
rice_yearly = rice_data.groupby('Year')['new production'].sum()
rice_yearly.plot(title='Rice Production Over Years')
```

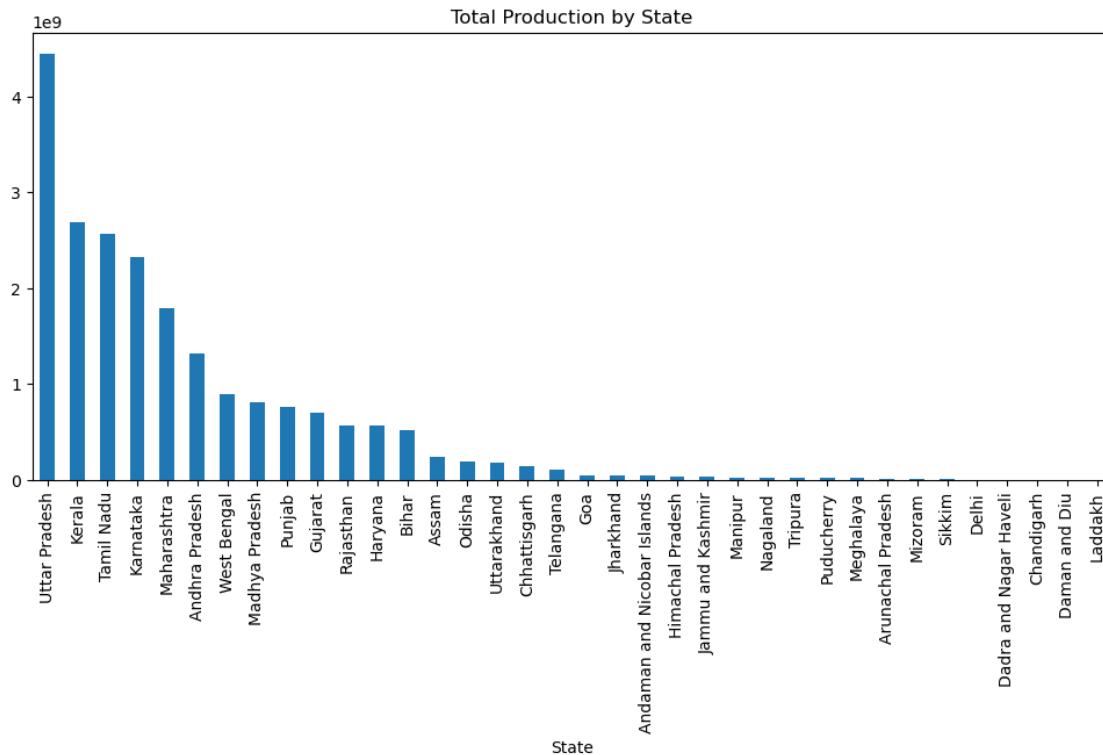
```
[248]: <Axes: title={'center': 'Rice Production Over Years'}, xlabel='Year'>
```



1.0.9 Geographical Trends

```
[249]: # Total production by State: "Which state has the highest total crop production
↳in the dataset overall?"
state_prod = agri_df.groupby('State')['new production'].sum().
↳sort_values(ascending=False)
state_prod.plot(kind='bar', figsize=(12, 5), title='Total Production by State')
```

```
[249]: <Axes: title={'center': 'Total Production by State'}, xlabel='State'>
```



1.0.10 Statistical Insight

```
[250]: agri_df.sample(4)
```

```
[250]:
```

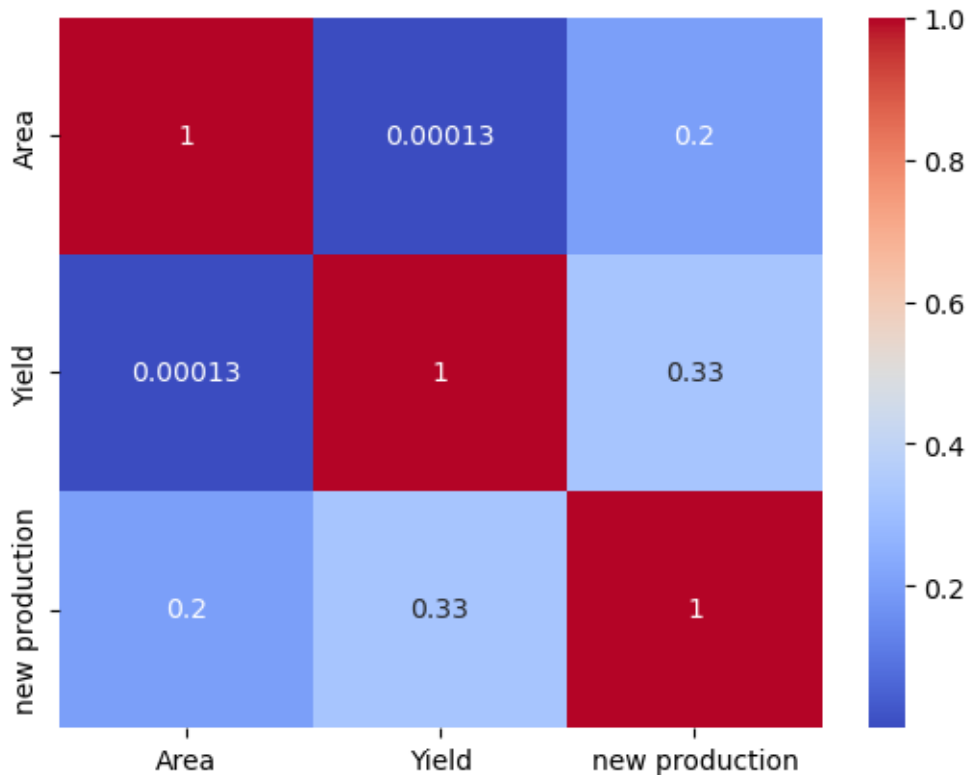
	State	District	Crop	Year	Season	Area \
26221	Nagaland	DIMAPUR	Sunflower	2001-02	Kharif	230.0
171245	Uttar Pradesh	KASGANJ	Onion	2012-13	Rabi	215.0
108785	Madhya Pradesh	CHHATARPUR	Turmeric	2009-10	Whole Year	106.0
328671	Odisha	DHENKANAL	Sugarcane	2000-01	Whole Year	554.0

	Area Units	Yield	new production
26221	Hectare	1.130435	260.0
171245	Hectare	23.004651	4946.0

```
108785    Hectare    0.603774         64.0
328671    Hectare   52.655054       29170.9
```

```
[251]: # Correlation
corr = agri_df[['Area', 'Yield', 'new production']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
```

```
[251]: <Axes: >
```



- Yield = production/Area
- Why Is Yield Important?

It helps compare how efficiently land is being used.

Two states may grow the same crop on the same area, but the one with higher yield is more productive (better soil, irrigation, seeds, etc.)

1.0.11 1. What crops are grown in a particular state?

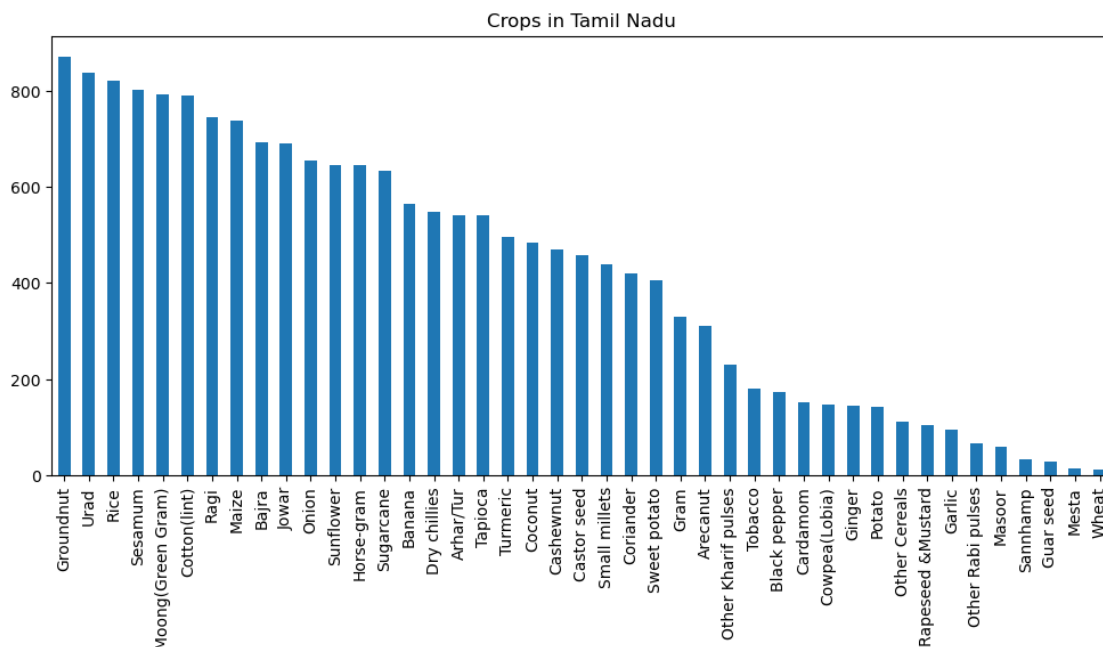
```
[252]: state = "Tamil Nadu"
crops_in_state = agri_df[agri_df['State'] == state]['Crop'].unique()
print(f"Crops grown in {state}:\n", crops_in_state)
```

Crops grown in Tamil Nadu:

```
['Arhar/Tur' 'Bajra' 'Banana' 'Black pepper' 'Cardamom' 'Cashewnut'
'Castor seed' 'Coconut' 'Coriander' 'Cotton(lint)' 'Dry chillies'
'Garlic' 'Ginger' 'Gram' 'Groundnut' 'Horse-gram' 'Jowar' 'Maize'
'Moong(Green Gram)' 'Onion' 'Other Kharif pulses' 'Potato' 'Ragi'
'Rapeseed &Mustard' 'Rice' 'Sesamum' 'Small millets' 'Sugarcane'
'Sunflower' 'Sweet potato' 'Tapioca' 'Tobacco' 'Turmeric' 'Urad' 'Wheat'
'Arecanut' 'Masoor' 'Cowpea(Lobia)' 'Other Rabi pulses' 'Other Cereals'
'Guar seed' 'Mesta' 'Sannhamp']
```

```
[253]: # Plot number of records(no of times) per crop.
crop_count = agri_df[agri_df['State'] == state]['Crop'].value_counts()
crop_count.plot(kind='bar', title=f"Crops in {state}", figsize=(12,5))
```

```
[253]: <Axes: title={'center': 'Crops in Tamil Nadu'}>
```



1.0.12 2. Which districts grow which crops in a state?

```
[254]: state = "Karnataka"
agri_df[agri_df['State'] == state].groupby('District')['Crop'].unique()
```

```
[254]: District
BAGALKOT      [Arecanut, Arhar/Tur, Bajra, Banana, Castor se...
BAGALKOTE     [Arecanut, Arhar/Tur, Bajra, Cashewnut, Castor...
BALLARI       [Arecanut, Arhar/Tur, Bajra, Cashewnut, Castor...
BANGALORE RURAL [Arecanut, Arhar/Tur, Banana, Black pepper, Ca...
```

BELAGAVI	[Arecanut, Arhar/Tur, Bajra, Cardamom, Cashewn...
BELGAUM	[Arecanut, Arhar/Tur, Bajra, Banana, Castor se...
BELLARY	[Arecanut, Arhar/Tur, Bajra, Banana, Castor se...
BENGALURU URBAN	[Arecanut, Arhar/Tur, Banana, Castor seed, Coc...
BIDAR	[Arhar/Tur, Bajra, Banana, Coconut, Coriander,...
BIJAPUR	[Arhar/Tur, Bajra, Banana, Castor seed, Coconu...
CHAMARAJANAGAR	[Arecanut, Arhar/Tur, Bajra, Banana, Cardamom,...
CHAMARAJANAGARA	[Arecanut, Arhar/Tur, Bajra, Black pepper, Cas...
CHIKBALLAPUR	[Arhar/Tur, Banana, Cashewnut, Castor seed, Co...
CHIKKABALLAPURA	[Arecanut, Arhar/Tur, Bajra, Cashewnut, Castor...
CHIKKAMAGALURU	[Arecanut, Arhar/Tur, Bajra, Black pepper, Car...
CHIKMAGALUR	[Arecanut, Arhar/Tur, Banana, Black pepper, Ca...
CHITRADURGA	[Arecanut, Arhar/Tur, Bajra, Banana, Castor se...
DAKSHIN KANNAD	[Arecanut, Banana, Black pepper, Cardamom, Coc...
DAKSHINA KANNADA	[Arecanut, Arhar/Tur, Bajra, Black pepper, Car...
DAVANGERE	[Arecanut, Arhar/Tur, Bajra, Banana, Black pep...
DHARWAD	[Arecanut, Arhar/Tur, Banana, Coconut, Coriand...
GADAG	[Arecanut, Arhar/Tur, Bajra, Banana, Castor se...
GULBARGA	[Arhar/Tur, Bajra, Banana, Black pepper, Casto...
HASSAN	[Arecanut, Arhar/Tur, Banana, Black pepper, Ca...
HAVERI	[Arecanut, Arhar/Tur, Bajra, Banana, Castor se...
KALABURAGI	[Arhar/Tur, Bajra, Castor seed, Coconut, Cotto...
KODAGU	[Arecanut, Banana, Black pepper, Cardamom, Coc...
KOLAR	[Arecanut, Arhar/Tur, Bajra, Banana, Castor se...
KOPPAL	[Arhar/Tur, Bajra, Banana, Castor seed, Coconu...
MANDYA	[Arecanut, Arhar/Tur, Banana, Black pepper, Ca...
MYSORE	[Arecanut, Arhar/Tur, Banana, Black pepper, Ca...
MYSURU	[Arecanut, Arhar/Tur, Bajra, Black pepper, Car...
RAICHUR	[Arhar/Tur, Bajra, Banana, Castor seed, Coconu...
RAMANAGARA	[Arecanut, Arhar/Tur, Banana, Cashewnut, Casto...
SHIMOGA	[Arecanut, Arhar/Tur, Bajra, Banana, Black pep...
SHIVAMOGGA	[Arecanut, Arhar/Tur, Bajra, Black pepper, Car...
TUMAKURU	[Arecanut, Arhar/Tur, Bajra, Black pepper, Car...
TUMKUR	[Arecanut, Arhar/Tur, Bajra, Banana, Black pep...
UDUPI	[Arecanut, Banana, Black pepper, Coconut, Dry ...
UTTAR KANNAD	[Arecanut, Arhar/Tur, Banana, Black pepper, Ca...
UTTARA KANNADA	[Arecanut, Bajra, Black pepper, Cardamom, Cash...
VIJAYAPURA	[Arecanut, Arhar/Tur, Bajra, Castor seed, Coco...
YADAGIRI	[Arhar/Tur, Bajra, Cashewnut, Castor seed, Coc...
YADGIR	[Arhar/Tur, Bajra, Castor seed, Coconut, Coria...

Name: Crop, dtype: object

```
[255]: state = "Karnataka"
district_crop_map = agri_df[agri_df['State'] == state].
    ↳groupby('District')['Crop'].unique()

for district, crops in district_crop_map.items():
```

```
print(f"{district}: {list(crops)}")
```

BAGALKOT: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Rapeseed &Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Turmeric', 'Wheat', 'Urad', 'Cashewnut', 'Tobacco', 'Cowpea(Lobia)', 'Black pepper', 'Ginger', 'Potato', 'Ragi', 'Tapioca']

BAGALKOTE: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Cashewnut', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

BALLARI: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Cashewnut', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rapeseed &Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

BANGALORE RURAL: ['Arecanut', 'Arhar/Tur', 'Banana', 'Black pepper', 'Castor seed', 'Coconut', 'Coriander', 'Dry chillies', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed &Mustard', 'Rice', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Turmeric', 'Urad', 'Bajra', 'Garlic', 'Cashewnut', 'Cowpea(Lobia)', 'Wheat', 'Cotton(lint)', 'Tobacco', 'Cardamom', 'Jowar', 'Mesta', 'Safflower', 'Linseed', 'Sannhamp', 'Soyabean']

BELAGAVI: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Cardamom', 'Cashewnut', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rapeseed &Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

BELGAUM: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed &Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Cashewnut', 'Ginger', 'Black pepper', 'Cowpea(Lobia)', 'Tapioca']

BELLARY: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Mesta', 'Cowpea(Lobia)', 'Soyabean', 'Cashewnut', 'Potato', 'Sweet potato', 'Tapioca']

BENGALURU URBAN: ['Arecanut', 'Arhar/Tur', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Dry chillies', 'Garlic', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Turmeric', 'Urad', 'Black pepper', 'Bajra', 'Cashewnut', 'Cowpea(Lobia)', 'Ginger', 'Linseed', 'Tapioca', 'Cotton(lint)', 'Soyabean', 'Tobacco', 'Wheat', 'Safflower', 'Sannhamp']

BIDAR: ['Arhar/Tur', 'Bajra', 'Banana', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Turmeric', 'Urad', 'Wheat', 'Black pepper', 'Cashewnut', 'Cowpea(Lobia)', 'Castor seed', 'Ragi', 'Tapioca', 'Tobacco']

BIJAPUR: ['Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Rice', 'Safflower', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Turmeric', 'Wheat', 'Soyabean', 'Urad', 'Cowpea(Lobia)', 'Potato', 'Arecanut', 'Ginger', 'Mesta', 'Ragi', 'Sannhamp', 'Tobacco']

CHAMARAJANAGAR: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Cardamom', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rice', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Tobacco', 'Turmeric', 'Urad', 'Ginger', 'Black pepper', 'Cashewnut', 'Soyabean', 'Tapioca', 'Wheat', 'Cowpea(Lobia)', 'Rapeseed & Mustard', 'Sannhamp', 'Linseed', 'Safflower', 'Sweet potato']

CHAMARAJANAGARA: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Black pepper', 'Cashewnut', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

CHIKBALLAPUR: ['Arhar/Tur', 'Banana', 'Cashewnut', 'Castor seed', 'Coconut', 'Coriander', 'Dry chillies', 'Gram', 'Groundnut', 'Horse-gram', 'Maize', 'Niger

seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Small millets', 'Sugarcane', 'Sunflower', 'Arecanut', 'Bajra', 'Cotton(lint)', 'Cowpea(Lobia)', 'Garlic', 'Ginger', 'Jowar', 'Peas & beans (Pulses)', 'Sweet potato', 'Turmeric', 'Linseed', 'Moong(Green Gram)', 'Safflower', 'Sesamum', 'Soyabean', 'Tapioca', 'Tobacco', 'Urad', 'Wheat']

CHIKKABALLAPURA: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Cashewnut', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

CHIKKAMAGALURU: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Black pepper', 'Cardamom', 'Cashewnut', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

CHIKMAGALUR: ['Arecanut', 'Arhar/Tur', 'Banana', 'Black pepper', 'Cardamom', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Cashewnut', 'Bajra', 'Cowpea(Lobia)', 'Soyabean', 'Linseed', 'Garlic']

CHITRADURGA: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Cashewnut', 'Potato', 'Cowpea(Lobia)', 'Black pepper', 'Cardamom', 'Ginger', 'Linseed', 'Sannhamp', 'Soyabean', 'Sweet potato']

DAKSHIN KANNAD: ['Arecanut', 'Banana', 'Black pepper', 'Cardamom', 'Coconut', 'Dry chillies', 'Ginger', 'Horse-gram', 'Moong(Green Gram)', 'Other Rabi pulses', 'Rice', 'Sesamum', 'Sugarcane', 'Sweet potato', 'Tapioca', 'Turmeric', 'Urad', 'Cashewnut', 'Gram', 'Maize', 'Cotton(lint)', 'Cowpea(Lobia)', 'Arhar/Tur', 'Bajra', 'Coriander', 'Garlic', 'Other Kharif pulses', 'Potato', 'Tobacco', 'Wheat', 'Groundnut']

DAKSHINA KANNADA: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Black pepper', 'Cardamom', 'Cashewnut', 'Coconut', 'Coriander', 'Cowpea(Lobia)', 'Jowar', 'Other Kharif pulses', 'Rice', 'Small millets', 'Sugarcane', 'Sweet potato', 'Tapioca', 'Tobacco', 'Wheat']

DAVANGERE: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Black pepper', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic',

'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Cashewnut', 'Cowpea(Lobia)', 'Cardamom']

DHARWAD: ['Arecanut', 'Arhar/Tur', 'Banana', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Tobacco', 'Urad', 'Wheat', 'Black pepper', 'Castor seed', 'Cashewnut', 'Bajra', 'Cowpea(Lobia)', 'Turmeric', 'Ginger', 'Sweet potato', 'Tapioca']

GADAG: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Peas & beans (Pulses)', 'Sannhamp', 'Soyabean', 'Sweet potato', 'Cowpea(Lobia)', 'Black pepper', 'Cashewnut', 'Mesta', 'Potato', 'Tapioca']

GULBARGA: ['Arhar/Tur', 'Bajra', 'Banana', 'Black pepper', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Ragi', 'Cowpea(Lobia)', 'Peas & beans (Pulses)', 'Arecanut', 'Cashewnut', 'Tapioca']

HASSAN: ['Arecanut', 'Arhar/Tur', 'Banana', 'Black pepper', 'Cardamom', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tobacco', 'Turmeric', 'Urad', 'Sannhamp', 'Cashewnut', 'Bajra', 'Cowpea(Lobia)', 'Tapioca', 'Safflower', 'Wheat']

HAVERI: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Cashewnut', 'Cowpea(Lobia)', 'Black pepper', 'Sweet potato', 'Cardamom', 'Tapioca']

KALABURAGI: ['Arhar/Tur', 'Bajra', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut',

'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Turmeric', 'Urad', 'Wheat']

KODAGU: ['Arecanut', 'Banana', 'Black pepper', 'Cardamom', 'Coconut', 'Dry chillies', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Maize', 'Other Rabi pulses', 'Other Kharif pulses', 'Ragi', 'Rice', 'Sweet potato', 'Tapioca', 'Tobacco', 'Arhar/Tur', 'Castor seed', 'Peas & beans (Pulses)', 'Potato', 'Sesamum', 'Turmeric', 'Cashewnut', 'Cowpea(Lobia)', 'Jowar', 'Sunflower', 'Coriander', 'Moong(Green Gram)', 'Small millets', 'Urad', 'Wheat', 'Bajra', 'Cotton(lint)', 'Onion', 'Sugarcane']

KOLAR: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Dry chillies', 'Gram', 'Groundnut', 'Horse-gram', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Small millets', 'Sugarcane', 'Sunflower', 'Urad', 'Cashewnut', 'Cowpea(Lobia)', 'Sesamum', 'Cardamom', 'Garlic', 'Ginger', 'Jowar', 'Sannhamp', 'Soyabean', 'Sweet potato', 'Tapioca', 'Turmeric', 'Cotton(lint)', 'Safflower', 'Tobacco', 'Wheat']

KOPPAL: ['Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Rice', 'Safflower', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Urad', 'Wheat', 'Mesta', 'Peas & beans (Pulses)', 'Potato', 'Rapeseed & Mustard', 'Sannhamp', 'Turmeric', 'Cowpea(Lobia)', 'Ginger', 'Cashewnut', 'Garlic', 'Ragi', 'Soyabean', 'Tobacco', 'Sweet potato', 'Tapioca']

MANDYA: ['Arecanut', 'Arhar/Tur', 'Banana', 'Black pepper', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rice', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Turmeric', 'Urad', 'Rapeseed & Mustard', 'Tapioca', 'Wheat', 'Cashewnut', 'Coriander', 'Cowpea(Lobia)', 'Tobacco', 'Bajra', 'Cardamom', 'Mesta', 'Safflower', 'Linseed', 'Sannhamp']

MYSORE: ['Arecanut', 'Arhar/Tur', 'Banana', 'Black pepper', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rice', 'Sesamum', 'Soyabean', 'Sugarcane', 'Sunflower', 'Tobacco', 'Turmeric', 'Urad', 'Rapeseed & Mustard', 'Tapioca', 'Cashewnut', 'Sweet potato', 'Cowpea(Lobia)', 'Garlic', 'Bajra', 'Linseed', 'Mesta', 'Small millets', 'Cardamom', 'Safflower', 'Sannhamp', 'Wheat']

MYSURU: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Black pepper', 'Cardamom', 'Cashewnut', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum',

'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

RAICHUR: ['Arhar/Tur', 'Bajra', 'Banana', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Sannhamp', 'Cowpea(Lobia)', 'Garlic', 'Mesta', 'Potato', 'Ragi', 'Soyabean', 'Sweet potato']

RAMANAGARA: ['Arecanut', 'Arhar/Tur', 'Banana', 'Cashewnut', 'Castor seed', 'Coconut', 'Dry chillies', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Turmeric', 'Urad', 'Bajra', 'Black pepper', 'Coriander', 'Cowpea(Lobia)', 'Garlic', 'Peas & beans (Pulses)', 'Tobacco', 'Cotton(lint)', 'Jowar', 'Potato', 'Cardamom', 'Safflower', 'Sannhamp', 'Soyabean', 'Wheat']

SHIMOGA: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Black pepper', 'Cardamom', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Safflower', 'Cashewnut', 'Potato', 'Coriander', 'Cowpea(Lobia)', 'Wheat', 'Sweet potato', 'Mesta', 'Sannhamp']

SHIVAMOGGA: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Black pepper', 'Cardamom', 'Cashewnut', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Onion', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rice', 'Sannhamp', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad']

TUMAKURU: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Black pepper', 'Cardamom', 'Cashewnut', 'Castor seed', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

TUMKUR: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Banana', 'Black pepper', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Peas & beans (Pulses)', 'Potato', 'Ragi', 'Rapeseed & Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat', 'Linseed', 'Cashewnut', 'Cowpea(Lobia)', 'Cardamom']

UDUPI: ['Arecanut', 'Banana', 'Black pepper', 'Coconut', 'Dry chillies',

'Ginger', 'Groundnut', 'Horse-gram', 'Moong(Green Gram)', 'Onion', 'Other Rabi pulses', 'Rice', 'Sesamum', 'Sugarcane', 'Sweet potato', 'Tapioca', 'Turmeric', 'Urad', 'Gram', 'Maize', 'Other Kharif pulses', 'Rapeseed &Mustard', 'Sunflower', 'Cashewnut', 'Cowpea(Lobia)', 'Cardamom', 'Coriander', 'Jowar', 'Tobacco', 'Wheat', 'Arhar/Tur', 'Bajra', 'Potato']

UTTAR KANNAD: ['Arecanut', 'Arhar/Tur', 'Banana', 'Black pepper', 'Cardamom', 'Coconut', 'Cotton(lint)', 'Dry chillies', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Ragi', 'Rapeseed &Mustard', 'Rice', 'Sannhamp', 'Sesamum', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Turmeric', 'Urad', 'Soyabean', 'Tobacco', 'Cashewnut', 'Cowpea(Lobia)', 'Bajra', 'Coriander', 'Potato', 'Small millets', 'Safflower']

UTTARA KANNADA: ['Arecanut', 'Bajra', 'Black pepper', 'Cardamom', 'Cashewnut', 'Coconut', 'Cotton(lint)', 'Cowpea(Lobia)', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rice', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tobacco', 'Turmeric', 'Urad']

VIJAYAPURA: ['Arecanut', 'Arhar/Tur', 'Bajra', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rapeseed &Mustard', 'Rice', 'Safflower', 'Sannhamp', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Tobacco', 'Turmeric', 'Urad', 'Wheat']

YADAGIRI: ['Arhar/Tur', 'Bajra', 'Cashewnut', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Garlic', 'Ginger', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Maize', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Rabi pulses', 'Other Kharif pulses', 'Potato', 'Ragi', 'Rice', 'Safflower', 'Sesamum', 'Small millets', 'Soyabean', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Tapioca', 'Turmeric', 'Urad', 'Wheat']

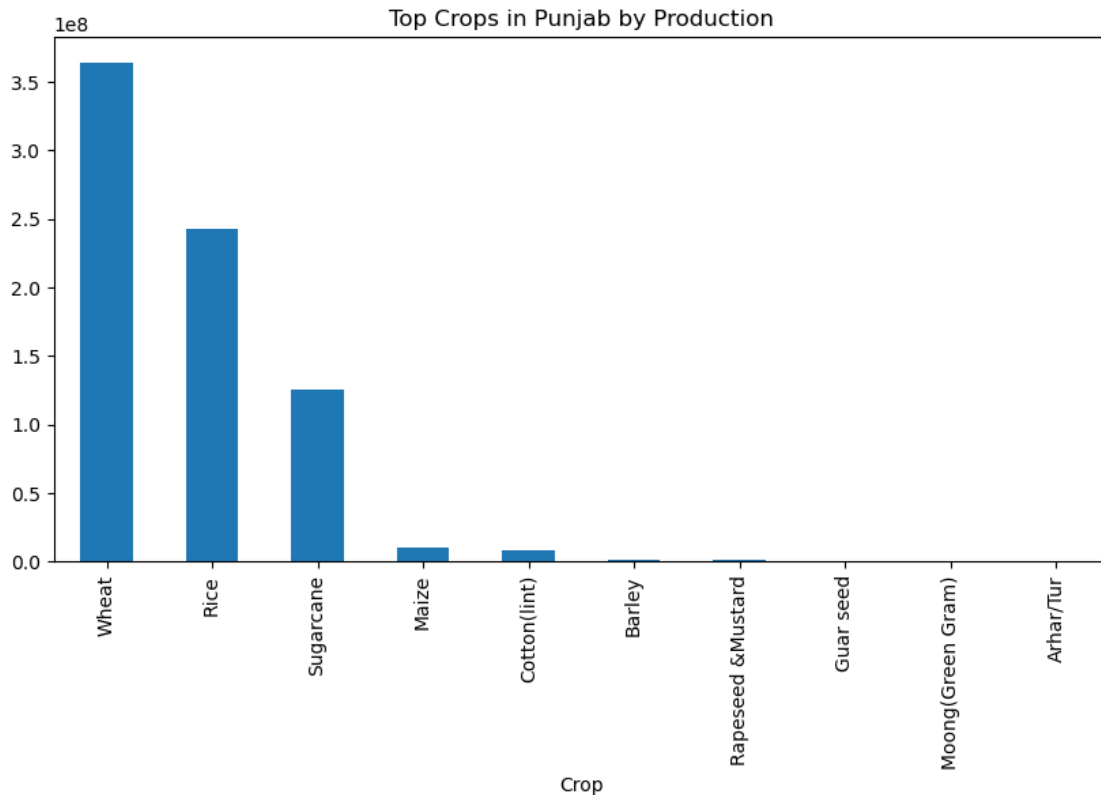
YADGIR: ['Arhar/Tur', 'Bajra', 'Castor seed', 'Coconut', 'Coriander', 'Cotton(lint)', 'Cowpea(Lobia)', 'Dry chillies', 'Gram', 'Groundnut', 'Horse-gram', 'Jowar', 'Linseed', 'Maize', 'Mesta', 'Moong(Green Gram)', 'Niger seed', 'Onion', 'Other Kharif pulses', 'Rapeseed &Mustard', 'Rice', 'Safflower', 'Sesamum', 'Small millets', 'Sugarcane', 'Sunflower', 'Sweet potato', 'Urad', 'Wheat', 'Cashewnut', 'Garlic', 'Other Rabi pulses', 'Potato', 'Ragi', 'Tapioca', 'Tobacco', 'Turmeric']

1.0.13 3. Top crops by production in a state

```
[256]: state = "Punjab"
#agri_df[agri_df['State'] == state].groupby('Crop')
top_crops = agri_df[agri_df['State'] == state].groupby('Crop')['new_
↳production'].sum().sort_values(ascending=False)
```

```
top_crops.head(10).plot(kind='bar', title=f"Top Crops in {state} by Production", figsize=(10,5))
```

[256]: <Axes: title={'center': 'Top Crops in Punjab by Production'}, xlabel='Crop'>

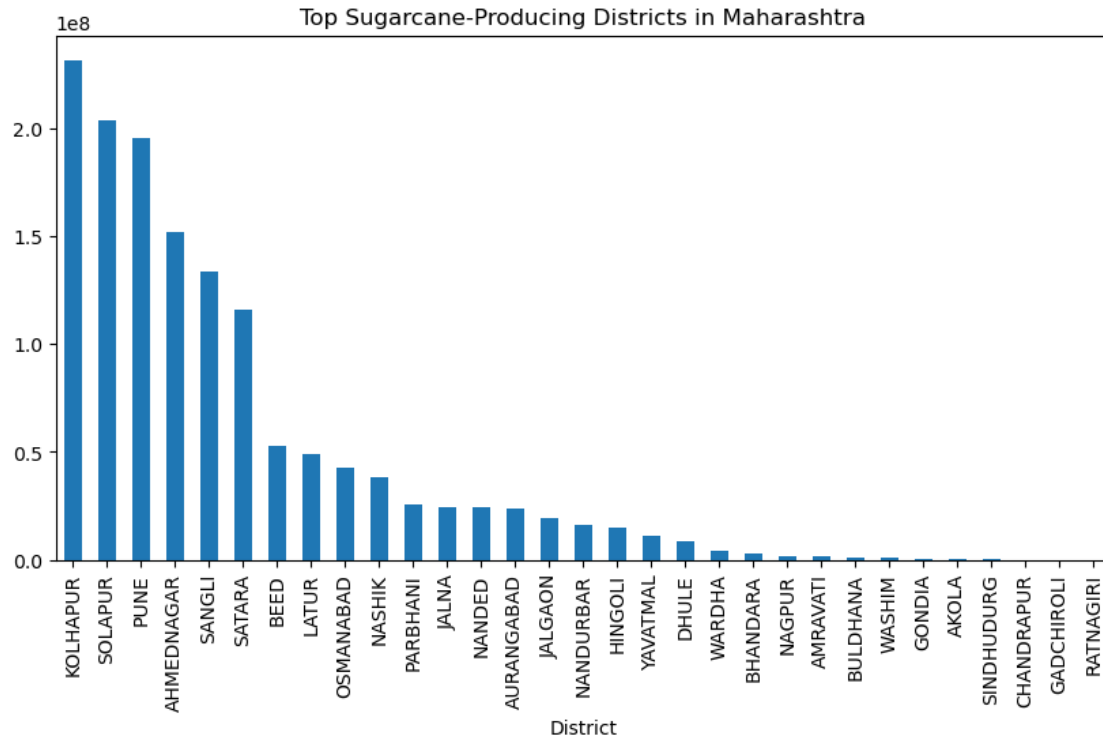


1.0.14 4. Top districts in a state for a specific crop

```
[257]: state = "Maharashtra"
crop = "Sugarcane"
top_districts = agri_df[(agri_df['State'] == state) & (agri_df['Crop'] == crop)] \
    .groupby('District')['new production'].sum() \
    .sort_values(ascending=False)

top_districts.plot(kind='bar', title=f"Top {crop}-Producing Districts in {state}", figsize=(10,5))
```

[257]: <Axes: title={'center': 'Top Sugarcane-Producing Districts in Maharashtra'}, xlabel='District'>

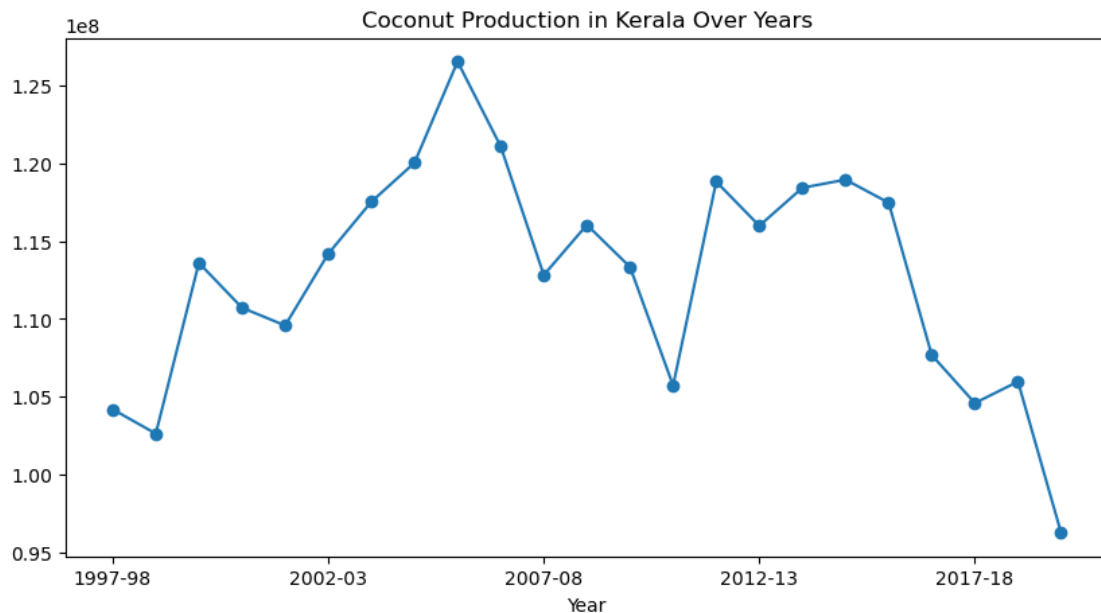


1.0.15 5. Year-wise trend of a crop in a state

```
[258]: state = "Kerala"
crop = "Coconut"
trend = agri_df[(agri_df['State'] == state) & (agri_df['Crop'] == crop)] \
        .groupby('Year')['new production'].sum()

trend.plot(marker='o', title=f"{crop} Production in {state} Over Years",
           figsize=(10,5))
```

```
[258]: <Axes: title={'center': 'Coconut Production in Kerala Over Years'},
       xlabel='Year'>
```



1.0.16 6. Seasonal crop patterns in a state

```
[259]: agri_df[agri_df['State'] == state].groupby(['Season', 'Crop']).size().
        ↪unstack().fillna(0)
```

```
[259]: Crop      Arecanut  Arhar/Tur  Banana  Black pepper  Cardamom  Cashewnut  \
Season
Autumn          0.0          0.0      0.0             0.0          0.0          0.0
Kharif           0.0          0.0      0.0             0.0          0.0          0.0
Rabi             0.0          0.0      0.0             0.0          0.0          0.0
Summer           0.0          0.0      0.0             0.0          0.0          0.0
Whole Year     322.0         21.0    308.0          322.0        140.0        322.0
Winter           0.0          0.0      0.0             0.0          0.0          0.0
```

```
Crop      Coconut  Cotton(lint)  Dry chillies  Garlic  ...  Small millets  \
Season
Autumn          0.0          0.0             0.0      0.0  ...          0.0
Kharif           0.0         13.0             0.0      0.0  ...         10.0
Rabi             0.0          0.0             0.0      0.0  ...          1.0
Summer           0.0          0.0             0.0      0.0  ...          0.0
Whole Year     322.0          4.0          15.0     20.0  ...          7.0
Winter           0.0          0.0             0.0      0.0  ...          0.0
```

```
Crop      Soyabean  Sugarcane  Sunflower  Sweet potato  Tapioca  Tobacco  \
Season
Autumn          0.0          0.0          0.0             0.0          0.0          0.0
```

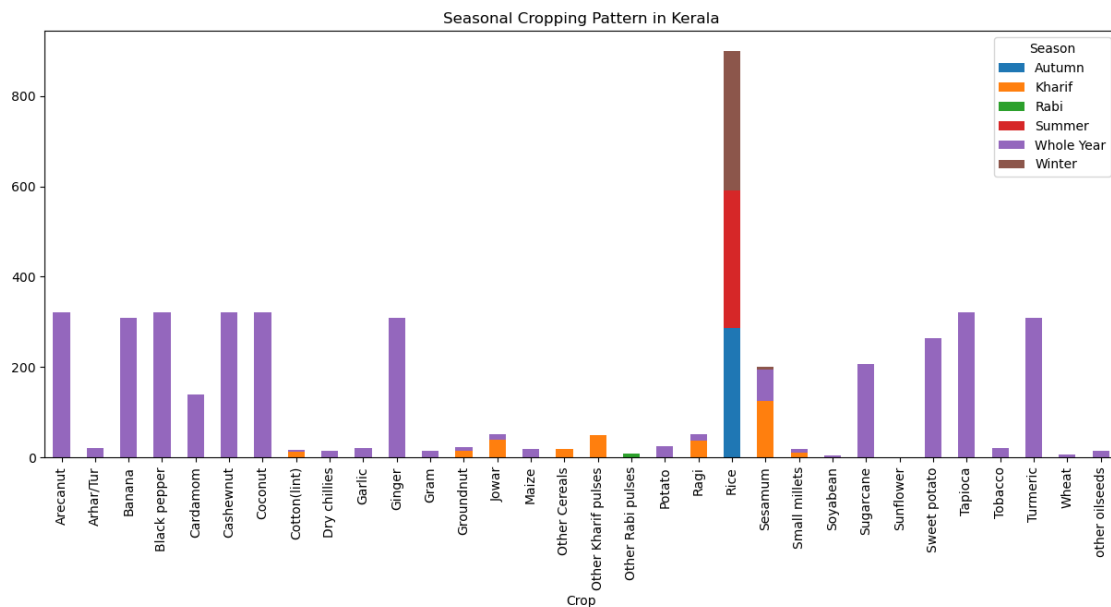
Kharif	0.0	0.0	0.0	0.0	0.0	0.0
Rabi	0.0	0.0	0.0	0.0	0.0	0.0
Summer	0.0	0.0	0.0	0.0	0.0	0.0
Whole Year	4.0	206.0	1.0	264.0	322.0	21.0
Winter	0.0	0.0	0.0	0.0	0.0	0.0

Crop	Turmeric	Wheat	other oilseeds
Season			
Autumn	0.0	0.0	0.0
Kharif	0.0	0.0	0.0
Rabi	0.0	0.0	0.0
Summer	0.0	0.0	0.0
Whole Year	308.0	6.0	14.0
Winter	0.0	0.0	0.0

[6 rows x 32 columns]

```
[260]: # Which crops are mostly grown in Kharif, Rabi, or Whole Year?
season_crop = agri_df[agri_df['State'] == state].groupby(['Season', 'Crop']).
    size().unstack().fillna(0)
season_crop.T.plot(kind='bar', stacked=True, figsize=(15,6), title=f"Seasonal
    Cropping Pattern in {state}")
```

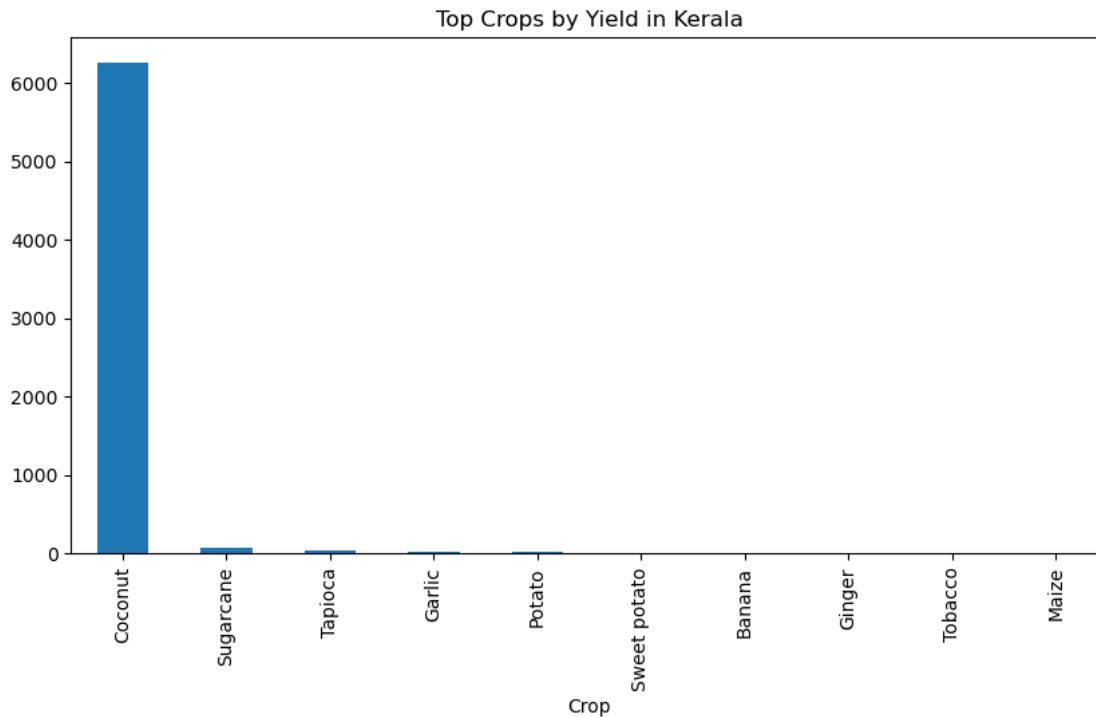
[260]: <Axes: title={'center': 'Seasonal Cropping Pattern in Kerala'}, xlabel='Crop'>



1.0.17 7. Compare yield across crops in a state

```
[261]: # Which crops have the highest yield in a state?
avg_yield = agri_df[agri_df['State'] == state].groupby('Crop')['Yield'].mean().
    ↪sort_values(ascending=False)
avg_yield.head(10).plot(kind='bar', title=f"Top Crops by Yield in {state}",
    ↪figsize=(10,5))
```

```
[261]: <Axes: title={'center': 'Top Crops by Yield in Kerala'}, xlabel='Crop'>
```



1.0.18 8. Crop Distribution Heatmap (State vs Crop) Analysis:

Visualize how widespread each crop is across different states.

```
[262]: import seaborn as sns
import matplotlib.pyplot as plt

# Create a matrix where each cell shows how many times a crop appears in a state
crop_state_matrix = agri_df.pivot_table(index='Crop', columns='State',
    ↪values='new production', aggfunc='count').fillna(0)

# Filter to top 20 crops (optional)
top_20_crops = crop_state_matrix.sum(axis=1).sort_values(ascending=False).
    ↪head(20).index
```

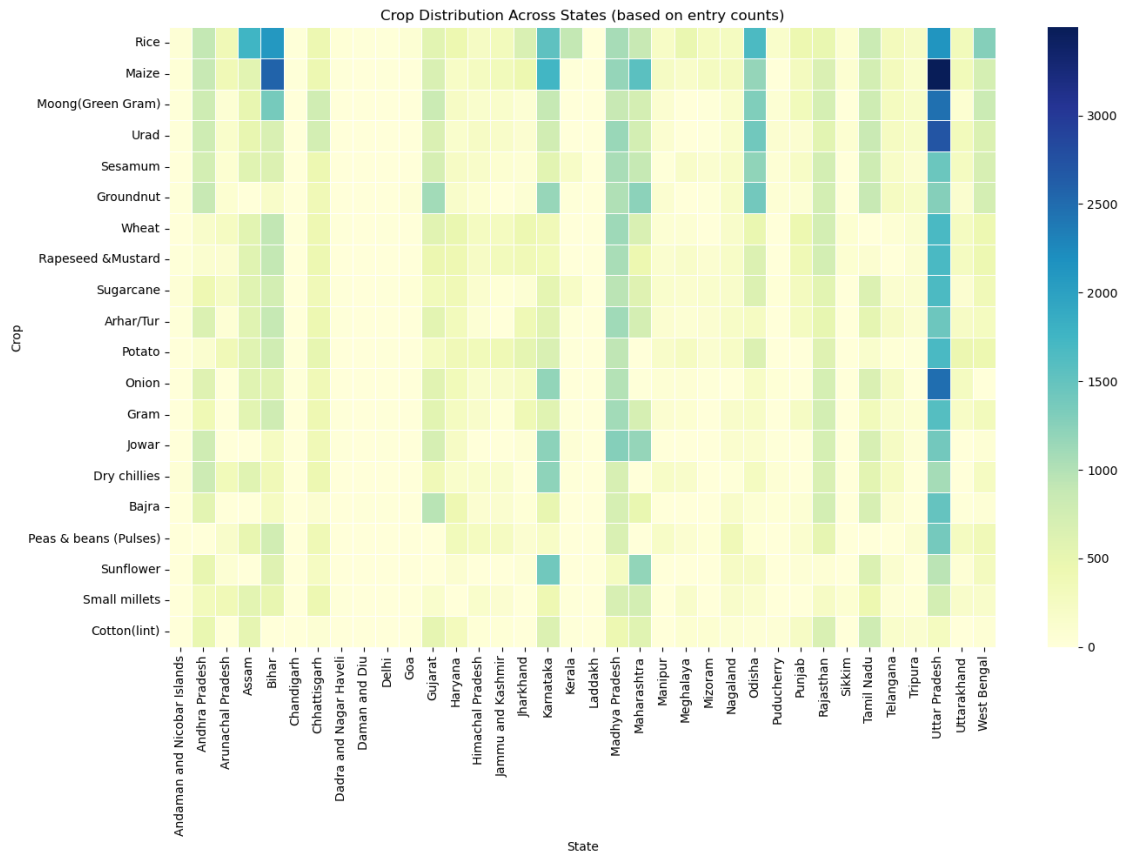


```

filtered_matrix = crop_state_matrix.loc[top_20_crops]

plt.figure(figsize=(14, 10))
sns.heatmap(filtered_matrix, cmap="YlGnBu", linewidths=0.5)
plt.title("Crop Distribution Across States (based on entry counts)")
plt.xlabel("State")
plt.ylabel("Crop")
plt.tight_layout()
plt.show()

```



1.0.19 9. Yield Comparison of the Same Crop Across States

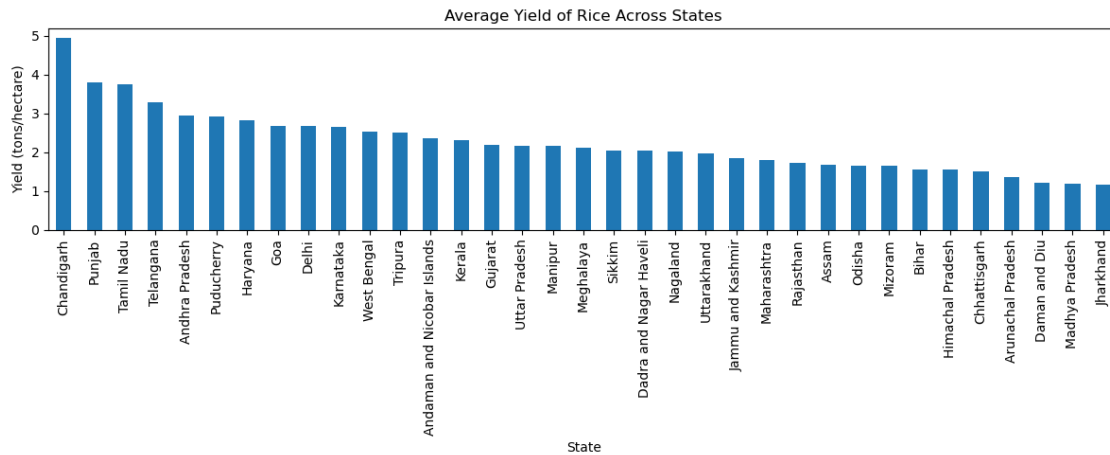
```

[263]: #Understand where a particular crop is grown most productively.
crop = "Rice"
rice_yield_by_state = agri_df[agri_df['Crop'] == crop].
    ↳groupby('State')['Yield'].mean().sort_values(ascending=False)

rice_yield_by_state.plot(kind='bar', figsize=(12, 5), title=f"Average Yield of_
    ↳{crop} Across States")
plt.ylabel("Yield (tons/hectare)")

```

```
plt.xlabel("State")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



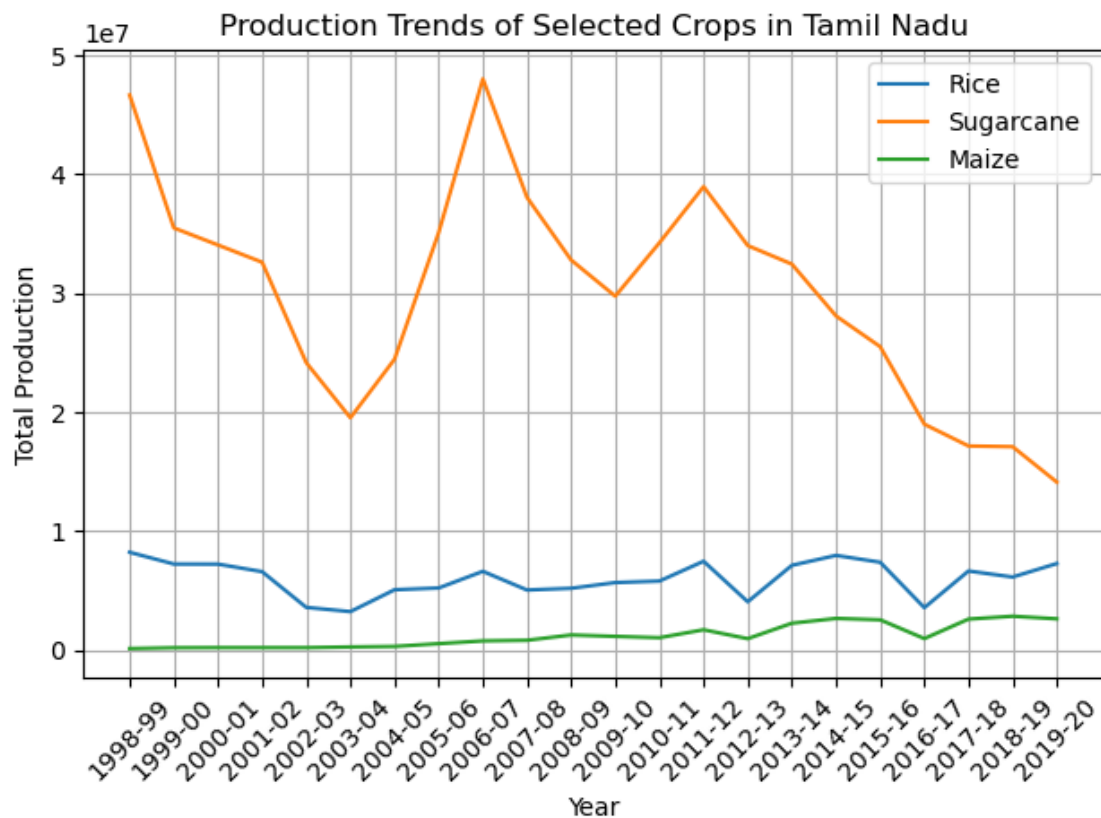
1.0.20 10. Time Series Comparison of Multiple Crops in One State Analysis:

Compare how different crops evolve over time in a state.

```
[264]: state = "Tamil Nadu"
crops = ["Rice", "Sugarcane", "Maize"] # choose your crops

for crop in crops:
    data = agri_df[(agri_df['State'] == state) & (agri_df['Crop'] == crop)]
    yearly = data.groupby('Year')['new production'].sum()
    plt.plot(yearly.index, yearly.values, label=crop)

plt.title(f"Production Trends of Selected Crops in {state}")
plt.xlabel("Year")
plt.ylabel("Total Production")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



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